



A remarkable rodent: Regeneration and reproduction in spiny mice (*Acomys*)

Ashley W. Seifert^{a,b,*} and Peter Temple-Smith^c

^aDepartment of Biology, University of Kentucky, Lexington, KY, United States

^bDepartment of Veterinary Anatomy and Physiology, University of Nairobi, Nairobi, Kenya

^cDepartment of Obstetrics & Gynecology, Monash University, Clayton, VIC, Australia

*Corresponding author: e-mail address: awseifert@uky.edu

Contents

1. Introduction	660
2. Regenerative biology	662
2.1 A chance encounter catalyzes a curiosity driven scientific adventure	662
2.2 A trapping horror resolves any doubt	666
2.3 Pseudoserendipity and the discovery of skin regeneration in wild <i>Acomys</i>	669
2.4 The model matures: Understanding complex tissue regeneration in a mammal	673
2.5 Potential for regeneration in other tissue and organ systems	679
3. Reproduction and developmental biology	681
3.1 Prologue	681
3.2 Menstruation and premenstrual tension: Pseudoserendipitous discoveries in female spiny mouse reproductive biology	683
3.3 Menopause and reproductive senescence	688
3.4 Reproduction in male spiny mice	689
3.5 Precocial development	692
3.6 Manipulating spiny mouse reproduction	694
4. Future perspectives	697
4.1 Community	697
4.2 Molecular resources	698
Acknowledgments	700
References	701

Abstract

Although certain organisms are chosen and employed to better understand a specific problem in biology (so-called model organisms), sometimes an animal model reveals its' biomedical importance by happenstance. In many ways, the advent of spiny mice (*Acomys*) as an emerging model to study regeneration and menstruation stands as a case study in scientific pseudoserendipity (Díaz de Chumaceiro, 1995). As we recount

in this chapter, the discovery of these phenotypes, while not entirely accidental, was nonetheless unexpected. In addition to recounting how we uncovered these unusual mammalian traits, we outline recent work by our groups and others that has begun to outline the cellular and genetic mechanisms underlying bonafide mammalian tissue regeneration and a human-like mode of reproduction in spiny mice.



1. Introduction

Every kind of organism has a special place in the epistemological sun.

E.O. Wilson

Relatively recent diversification among murid species provides a strong context for comparative studies seeking to understand the genomic basis for speciation as well as the cellular and genetic basis for lineage and clade-specific traits. Despite these advantages, other than *Mus*, *Rattus* and a handful of other species, murid rodents remain woefully understudied from the standpoint of developmental and reproductive biology.

Commonly referred to as spiny mice, *Acomys* are an enigmatic genus of rodents that are taxonomically grouped with *Lophuromys*, *Deomys* and *Uranomys* in the subfamily Deomyinae (Alhajeri, Hunt, & Stepan, 2015). Together with the Gerbillinae, Murinae and the single species clade Lophiomyinae, these four subfamilies represent the most diverse group of mammals on earth (Stepan & Schenk, 2017). This is all the more impressive given the relative youth of the murid clade (~18mya) and that radiations in these three multispecies subfamilies occurred within the last 10 million years (Fig. 1) (Stepan & Schenk, 2017).

Among murids, *Acomys* stand alone in possessing a suite of unique biological traits that are uncommon even among rodents (e.g., precocial young, highly social group dynamics, flexible sleep patterns, etc.) and several reviews have covered these topics in detail (Brunjes, 1990; D'Udine & Alleva, 1988; Houghton, Gawriluk, & Seifert, 2015; Shafrir, Ziv, & Kalman, 2006). Adding to this list of unique murid traits found in *Acomys*, work by our respective groups uncovered two additional phenotypes: complex tissue regeneration and a human-like pattern of reproduction (Bellofiore et al., 2017; Bellofiore, Rana, Dickinson, Temple-Smith, & Evans, 2018; Gawriluk et al., 2016; Seifert et al., 2012). As our respective discoveries have gained attention among biologists and biomedical researchers, a growing community of researchers has begun to embrace *Acomys* as a useful model to study these and other traits.

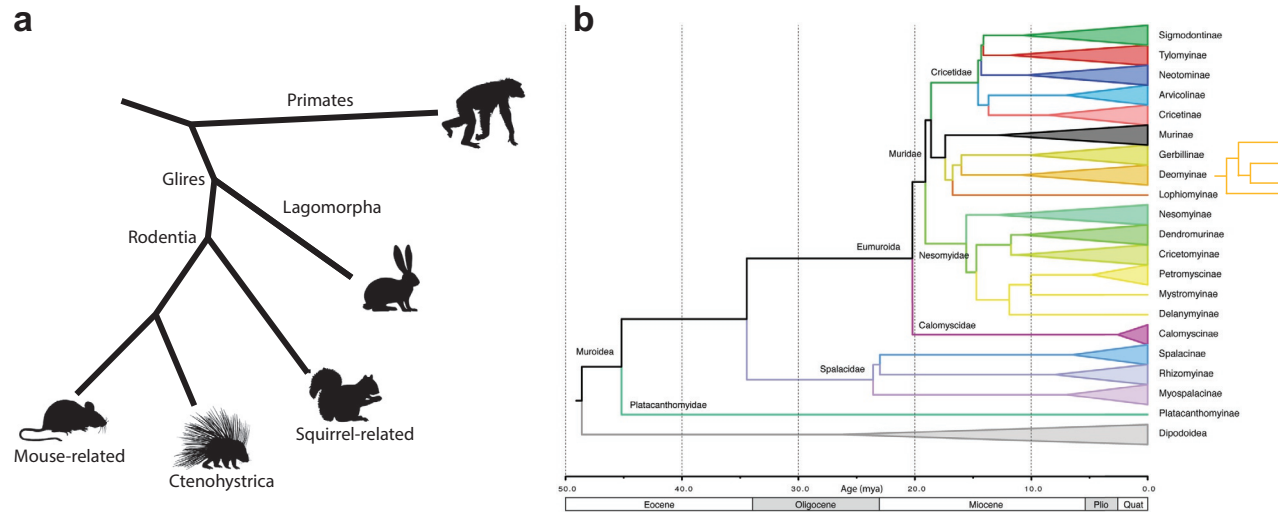


Fig. 1 Phylogenetic summary of murid subfamilies and taxonomic relationships of *Acomys*. (A) General relationship between primates, lagomorphs and rodents where all three groups shared a common ancestor approximately 90 million year ago (timetree.org). (B) Maximum-likelihood chronogram depicting taxonomic relationships among the major muroid subfamilies. *Acomys* are grouped with *Lophuromys*, *Deomys* and *Uranomys* in the subfamily Deomyinae. The Deomyinae, Gerbillinae, Murinae and the single species clade Lophiomyinae, comprise the Muridae, the largest family of rodents and mammals. General taxonomic relationships in (A) based on a detailed phylogenetic reconstruction of rodents inferred from a molecular supermatrix based on 11 mitochondrial and nuclear genes (Fabre, Hautier, Dimitrov, & Douzery, 2012). Panel (B) adapted from Steppan, S. J. and Schenk, J. J. (2017). Muroid rodent phylogenetics: 900-species tree reveals increasing diversification rates. *PLoS One* 12, e0183070. under Creative Commons Attribution License (CC BY 4.0).

August Krogh opined that one or several animals exist to conveniently study any scientific problem. He also lamented these animals would remain unknown to the persons most needing them and he encouraged scientists to “...*apply to the zoologists to find them and to lay our hands on them*” (Krogh, 1929). Writing years later on the importance of returning to taxon-centered research, E.O. Wilson wrote that the so-called inverse Krogh rule applied with equal force and that “...*for every organism there exists a problem for the solution of which it is ideally suited and others for which it is useless*” (Wilson, 1989). As adherents to approaches that some might consider antagonistic, we believe both strategies expand the horizon for scientific discovery and in doing so, unshackles those toiling away as they practice the progression of natural science.



2. Regenerative biology

Discovery is to see what everybody else has seen and think what nobody else has thought.

Albert Szent-Györgyi, from his book, Bioenergetics

These little guys jump right out of their skin!

Truman Young

2.1 A chance encounter catalyzes a curiosity driven scientific adventure

Whether at meetings or in casual conversation, scientists and nonscientists frequently ask to know the story of how I (Ashley) came to study regeneration in *Acomys*. Depending on the context, I usually recount an abridged version of how I traveled to Kenya and ultimately discovered their regenerative abilities. Devoid of decorative hyperbole, the complete story presents either an impassioned defense for pursuing curiosity driven science or a cautionary tale of the career risks associated with this approach. Whichever lesson one derives, much like the observation of female menstruation in these animals, both remind us that sometimes discoveries lie hidden in plain sight.

The spiny mouse regeneration story was born in the Biology department at the University of Florida. Nearing the end of my doctoral studies, I had chosen to remain in Gainesville to pursue postdoctoral research. Although staying at one's doctoral institution is generally frowned upon when pursuing a career in academia, two factors were prominent in my decision to stay in Gainesville, and both are pertinent to the story which follows. First, as I mulled over options for choosing a postdoctoral lab, my partner was in the

midst of graduate school and my heart was intent on remaining with her until she completed her dissertation research. Second, although there were many biological problems intriguing enough to consider studying as a postdoc, none were as seductive to me as trying to understand interspecific variation in regenerative ability (i.e., depending on the species, injury elicits either fibrotic repair or regenerative healing in response to tissue damage). Fortunately, Malcolm Maden, a prominent regeneration biologist, had recently moved his lab from King's College, London to the University of Florida. As a member of my thesis committee, we were already acquainted and sitting together in his office one afternoon, I made an impassioned pitch to join his lab to pursue regeneration research. While he was delighted to have me join his group, we needed to come up with a way to support my salary. Luckily, fate intervened in the form of an opening on a training grant and the die was cast. Heart and head aligned, I was thrilled to have my future postdoc plans settled.

With my dissertation defense looming, I decided that a well-deserved break would be in order before I launched into my postdoc research. What better way to celebrate defending my thesis than traveling to Kenya to help my partner Megan collect field data for her own dissertation. As an ecologist studying how plants defend themselves from herbivores, she was set to start her field season at Mpala Research Centre in Laikipia. Naturally, Megan was excited to have an extra pair of hands in the field and I was more than happy to lend them. Of course, there was the small matter of paying for my trip that needed to be settled before I went anywhere. Little did I know, a chance encounter was about to trigger a chain of events that would put me on a plane to Kenya and set the course for my postdoctoral research.

Around the time I had settled my postdoctoral plans, Truman Young, a community ecologist from UC Davis who had spent his career working at Mpala, had been invited to our department to give a seminar. Truman was Megan's academic grandfather and the evening after his talk we struck up conversation at the post-seminar social. After introducing myself, I shared my post-graduation plans to spend several months at Mpala with Megan. Not knowing that I was a developmental biologist, he asked me what I planned to work on during my postdoc. After a bit of back and forth, I started describing how species can heal injuries quite differently and that for my postdoc I planned to find out why. At some point, the conversation pivoted to the concept that regeneration might be an adaptive trait and I explained that this idea had a rich history which had formed the basis for

a contentious debate between Thomas Hunt Morgan and August Weisman (Esposito, 2013). Like Morgan, I believed (and still do) that the distribution of animal regeneration could not be explained by purely adaptive means.

Before I could continue, he stroked his mustache and blurted out something about “these mice in Kenya” and then suggested that these rodents might be relevant to my larger research question. “When you handle them, these little guys jump right out of their skin,” he said. Dumfounded, I asked him what he meant. He recounted a decades-old conversation with Steve Takata, a graduate student working at Mpala who had trapped spiny mice and who had observed what he described as a skin shedding behavior. Truman recounted how supposedly routine handling would result in massive skin tears and in the most extreme cases, near complete degloving. The working hypothesis was that it could act as predator defense mechanism, that is, if the mouse survived its injuries. “When you go over to Kenya this summer why don’t you trap some? Maybe they regenerate their skin?”

I thought about the conversation for the rest of the night. Why would a mammal evolve skin that could tear so easily and, in the process, readily expose themselves to infection? Losing a tail sheath or an entire tail was one thing, and many small mammals were known to exhibit tail loss (Shargal, Rath-Wolfson, Kronfeld, & Dayan, 1999), but losing large pieces of skin was an entirely different matter. My mind was filled with questions. How could these rodents maintain water balance? How did they fight infection? How did they heal their wounds? *Did* they heal their wounds? While the skeptic in me thought the behavior implausible, the dreamer in me found the potential to uncover a biological explanation seductive.

If there was credence to this behavior, then someone must have published an account somewhere. Over the next several weeks, I poured through primary research papers and also dipped into the gray literature world of technical reports and theses. My efforts turned up nothing beyond the occasional account of tail loss. I reached out to Megan’s advisor, Todd Palmer, who had overlapped with Steve Takata during his own dissertation field work. He didn’t remember conversations about skin shedding, so he sent out an email inquiry to other researchers working at Mpala. Later that week he forwarded me a picture taken by Jake Goheen, a mammalogist working at Mpala, that showed a spiny mouse missing a large piece of back skin. It appeared that Truman’s memory was spot on.

Soon enough I had outlined what seemed like a straightforward plan: go to Kenya, catch spiny mice, investigate skin tearing and if possible, characterize wound healing. With the plan hatched, all I needed to do was find a

way to pay for it! With no data to speak of except for a single picture, I found whatever fellowships I could and applied for money (Explorers Club, Lewis and Clark Foundation, National Geographic). Given the fellowship timelines, at best they would reimburse the project, although regrettably none panned out. A few weeks later I pitched my project idea to Todd. I reckoned that one way or another, characterizing what was happening with spiny mice in the field would make for an interesting manuscript. In exchange for authorship, I asked for some logistical support in the field and two thousand dollars to cover my plane ticket and to offset my accommodations. To my surprise, he agreed to this arrangement and with help from Megan, I set about the task of lining up research permits to trap rodents in Kenya.

Having previously conducted research in Uganda, I acutely understood the problems associated with working in East Africa. Starting a project from scratch in Africa involves all kinds of logistical hurdles in addition to the normal challenges of simply carrying out the research. While Todd and Mpala Research Centre could provide some of what I needed in the field, trapping mice that no one actively worked with would prove to be problematic. As I slowly worked to line up the necessary Kenyan permits from afar, two more introductions would facilitate key components necessary for the proposed experimental work.

The first introduction concerned the wound healing experiments I hoped to complete. Luckily, my advisors introduced me to the developmental biologist and wound healing expert Paul Martin. My graduate advisor Martin (Marty) Cohn and Malcolm were very well acquainted with Paul, and with no expertise in wound healing between the three of us, we arranged a call so I could discuss my ideas with Paul. Several months before heading into the field, Paul and I connected by phone in Marty's office. After explaining the basis for my expedition, Paul offered insight into the most appropriate paradigm for a pilot wound healing experiment and was happy to answer my endless barrage of questions. I wanted to be as prepared as possible in case I was able to trap enough animals. Before the conversation ended, I asked him what he thought about the potential for skin regeneration in spiny mice given their apparent propensity for skin autotomy. He responded by proposing a bet that the likely outcome of their skin injuries would be normal fibrotic repair (not regeneration), although he conceded they might heal faster. Whatever the outcome, he wished me well on my expedition and told me to look at their inflammatory response if things got interesting.

The second introduction concerned collaborative support from within Kenya. Despite the viewpoint that science knows no borders, the fact

remains that foreign scientists are guests. Permits aside, evolving national restrictions on visiting scientists are a recent development stemming from a historical legacy of exploitation and economic abuse, real or perceived. While I hoped to establish connections with Kenyan scientists once I arrived, another chance encounter provided an introduction that established a close collaboration, and friendship that I enjoy to this day. Ketan Patel, a British muscle biologist, had recently visited Marty while on holiday in the States. Ketan and Marty had been at the University of Reading together before Marty joined the Biology Department at Florida and Ketan and Malcolm were friends from the developmental biology community in England. One day in the lab, I mentioned to Ketan my pending trip to Kenya. As it turned out, one of his former students had graduated from the veterinary medicine program at the University of Nairobi and still had contacts there. After several emails, I was introduced to Stephen Kiama, then Chairman of the Department of Veterinary Anatomy and Physiology at the University of Nairobi. Professor Kiama had a longstanding interest in monocytes dating back to his own dissertation research in Bern and he found the nascent spiny mouse project particularly intriguing. With my departure weeks away, we arranged to meet in Nairobi when I landed to discuss project logistics. With seemingly all my ducks in a row, I successfully defended my thesis and packed my bags for the field.

2.2 A trapping horror resolves any doubt

When I arrived in Nairobi, I promptly made my way to the University of Nairobi's Chiromo Campus. Professor Kiama and I clicked right away. We leisurely exchanged personal histories, chatted about our research interests and even managed to wade into a bit of national politics. The Department of Veterinary Anatomy and Physiology had all the hallmarks of a classical Zoology department and walking the corridors I felt perfectly at home. Down the hall, a large theater ringed with skeletons stood idle for a laboratory in Comparative Anatomy. At the rear of the building stood an old animal house separated from the main building by a courtyard. Inside one of the rooms was a small colony of naked mole rats. Another room concealed a group of pancake tortoises (*Malacochersus tomieri*) in large sandboxes.

Later that morning Kiama introduced me to one of his graduate students, John Kimani, who showed me around the department. Besides the animals themselves, the inhalant anesthetic halothane was the most critical item I needed to procure if I was to perform any wound healing experiments.

Luckily, John helped me acquire a bottle of halothane from a pharmacy in the Central Business District; a half-day affair that introduced me to Nairobi's notorious traffic. Everything, it seemed, was falling neatly into place.

The next day, we were off to Laikipia for 2 months. Despite the aforementioned string of coincidences, introductions, and a bit of good luck, I was fully prepared for any number of small disasters which might occur in the field. What I didn't predict was an abdication of support from an American researcher who had promised me access to his Sherman traps and tips on trapping *Acomys*. When I arrived at Mpala, he needed the traps for himself and honestly had no idea what type of habitat *Acomys* preferred. I was beside myself. Disappointed as I was, there was no use wallowing in my misfortune. Instead, I spent days in the field with Megan measuring *Acacia* branches and tending to a seemingly endless array of spine punctures through my skin. As days turned into weeks, I immersed myself in her work while I soaked in one of the most amazing landscapes on Earth. On off days, I built wooden cages to house animals I didn't have and waited, albeit a bit impatiently, for the possibility of getting Sherman traps. Perhaps I *was* nuts for assuming a project I had spent months arranging with absolutely no experience in small mammal trapping or wound healing would simply work out because I wanted it too. Then, after 3 weeks of badgering and pleading, I finally acquired 50 Sherman traps.

At first, trapping *Acomys* seemed impossible. The savannah was expansive, and with 50 traps I wasn't catching much of anything. The spiny mice that researchers did catch in small mammal surveys were few and far between and comprised a very small percentage of the endemic species diversity. Late one afternoon after lamenting my inability to trap *Acomys*, I was reminded that Steve Takata had worked on kopjes (Fig. 2). Kopjes are very old rock

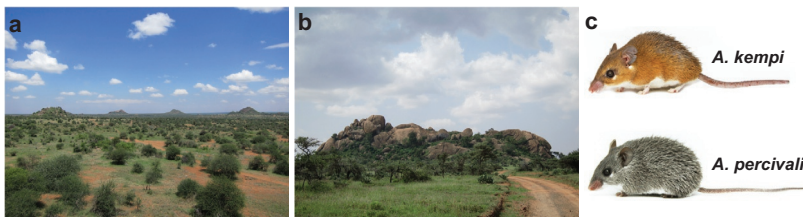


Fig. 2 *A. kempfi* and *A. percivali* inhabit rock outcroppings in Kenya. (A) Rock outcroppings called kopjes dot the savannah landscape in Laikipia county, Kenya. (B) A single kopje emerging from the red soil, acacia dominated savanna. (C) *A. kempfi* and *A. percivali*. *A. kempfi* is notable for the rufous color of hairs covering its dosum. *A. percivali* retains a gray coat throughout adulthood.

formations that dot the landscape like islands (Fig. 2). That night I set 50 traps around, and on, a kopje named Jessel's Rock at the border of Peter Jessel's Farm and Mpala. When I returned the next morning, all the traps that we had set next to the kopje were empty. However, of the 25 traps we set *on* the kopje, I could already see that several trap doors were closed. Among these, several contained a small, slate gray rodent with a white belly and spiny hairs covering its entire body. This was *Acomys percivali*, Percival's spiny mouse (Fig. 2) (Kingdon, 1984). As I removed that first spiny mouse from the trap, a horror ensued. Partly because I had almost no experience trapping rodents and partly because I was nervous, I awkwardly grabbed the spiny mouse in the middle of its back. The animal used its legs to push off my hand and a piece of skin was torn from its back. As the spiny mouse fell to the ground and took off running into the rocks, I stared down at a little piece of skin in my hand. I couldn't believe it. The spiny mouse had escaped by tearing off its own skin.

That night I set my traps on the kopje again. Returning the next morning, I once again found that many of my traps were occupied. This time, I simply kept the spiny mice I captured inside the traps until releasing them into a large wooden enclosure I had found back at the field station. In addition to *A. percivali*, I had also trapped *Acomys kempi*, a rufous colored species that was slightly larger by comparison, but undoubtedly spiny. This was Kemp's spiny mouse (Fig. 2). Over the next several weeks, I decided to set my traps exclusively on kopjes. In this manner, I figured out that *Acomys* preferred rocky habitat and that they were not the least bit trap shy. In a matter of weeks, I had already captured as many animals as I could hold in the compartmentalized wooden boxes I had constructed. With time remaining, I could now focus on the wound healing experiments I had hoped to conduct.

Observing and working with the animals at Mpala, I soon determined that their skin was uniformly weak. While I would later quantify the skin's tensile strength (Seifert, Kiama, et al., 2012), for now it was obvious that the skin was abnormally weak. I also followed animals with variable sized wounds and noticed that the skin appeared to heal on a typical time course (Clark, 1996). Moreover, being loose-skinned mammals, contraction from the wound margins accounted for most of the wound closure I observed just as Paul had predicted. Given the limited duration of my trip, I hadn't been able to complete anything more than a cursory assessment of the healing process in small, irregular wounds. Time was running out so I collected what samples I could to bring back to the States. I had, however, transported

50 animals back to the animal house in the Department of Veterinary Anatomy and Physiology in the hopes I would return. In the meantime, I figured that John could attempt to breed them to gather information on their reproduction.

Back in Malcolm's lab at Florida, a study describing fragile skin in geckos became a template for my histological investigations of the unwounded skin (Bauer, Russell, & Shadwick, 1989). Geckos have a fracture plane in their dermis that allows them to lose and rapidly regenerate the upper dermis in response to autotomy. However, there was absolutely no evidence for this in spiny mouse skin. In retrospect, it was not surprising because the natural injuries I observed appeared to be full-thickness defects. Unfortunately, my wound healing experiments had gone poorly, and I returned with only a handful of tissue samples from relatively early time points. Based on wound healing studies in mice, this would still provide important material to look at tissue inflammation, but with the material I had it was impossible to systematically address the long-term outcome of tissue loss and healing (Mori, Shaw, & Martin, 2008). After completing what I considered a proof-of-principle study, tissue regeneration was far from my mind. For the uninitiated, wound healing is messy business, and from all the papers I read, the outcome in mammals was always the same. In fact, as I tunneled deeper into the wound healing and regeneration literature, it became apparent that with the exception of fetal wound healing models, no one had described an adult skin healing model where the outcome was regenerative healing. This revelation shifted my attention to investigating skin healing in salamanders (Seifert, Monaghan, Voss, & Maden, 2012), although I was still very much committed to tying up the spiny mouse project into something publishable.

2.3 Pseudoserendipity and the discovery of skin regeneration in wild *Acomys*

If Truman was the catalyst for the spiny mouse project, Malcolm was the fuel. Rather than discourage my curiosity, Malcolm had actively encouraged it. During my first summer, he joined me for several weeks in Kenya to see the animals for himself and to help me with the wound healing experiments. Working side-by-side together had validated my belief that young scientists should be encouraged to study what captivates them regardless of the associated career risks. When no organizations would fund my project proposals to return the following summer, Malcolm provided research funds to help support the work. I had mapped out a plan to complete two components of

the project which I thought would help me wrap up the spiny mouse story. First, I would quantify skin weakness using a tensometer at the University of Nairobi. Second, I would complete a rigorous assessment of wound healing that included deep time points to assess the full extent of skin repair. Although the results would likely end up being far more descriptive than explanatory, a paper would bring this interesting behavior to the broader scientific community. The following summer I spent 2 months in the field and 1 month in Nairobi. While new adventures and disasters were plentiful, I managed to find a mechanical engineer in Nairobi who could help me record tensile strength from skin samples. Equally important, I was able to collect tissue samples that would allow me to completely characterize the wound repair process. With a full parcel of tissue samples in hand, I headed back to Florida for bench work.

Although I had observed remarkable healing in the spiny mice, my general skepticism to a healing outcome other than fibrotic repair obscured the notion of regeneration. For the wound healing experiments, I settled on a full-thickness, excisional model (1.5 cm diameter wounds). Gross observation of healing skin wounds in Kenya appeared to confirm my general expectation that contraction would dramatically shrink the wound bed which would later fill with scar tissue to complete repair in the center.

When I finally settled into a dissecting scope to prepare my tissue samples, I worked through the tubes in a temporal sequence. Uninjured skin followed by early healing time points and finally my 4-week samples. As I carefully emptied each parafilm wrapped tube into a petri dish of saline, my early time points were typical with a barren wound bed surrounded by clipped hairs at the wound margins. When I began to look at samples collected 4 weeks after injury, I couldn't believe my eyes. All throughout the wound bed I could see the pigmented tips of large spiny hairs erupting through the epidermis alongside much thinner guard hairs (Fig. 3). Cautiously, I leaned back in my chair and chuckled. The wound bed I was looking at didn't resemble the wound induced hair follicle neogenesis observed by some investigators when very large wounds were made in mice (termed WIHN). In those instances, new hairs emerged in small bunches from the center of the wound and were separated from the margins by a ring of scar tissue (Wang et al., 2015). Cutting parallel to the hairs, I could clearly see two hair types extending from the margins across the entire area (Fig. 3). What had apparently occurred in these skin wounds was regenerative healing instead of fibrotic repair! Although the discovery was not purely accidental, it was definitely an unexpected result. To embrace it was another



Fig. 3 *De novo* hair follicle regeneration in *Acomys*. (A) A piece of wounded back skin recovered 28 days after the original 1.5 cm diameter, full thickness skin injury. Large, spiny hairs in unwounded skin are visible outside the wound margins (WM) and the tips of newly generated hairs can be seen emerging from within the wound bed. (B) Longitudinal cut through wound bed center showing regenerated spiny (awl) and guard hairs across the entire bed. Dotted lines depict the wound margins (WM).

matter altogether. As the gravity of the discovery slowly sunk in, it was replaced with a mixture of excitement and doubt. Now I would need to convince myself that the spiny mice were indeed doing something remarkable. I would spend the rest of the year trying to do just that.

As I set about working up my dataset, I read every published report I could find pertaining to regeneration in mammals. This had the intended consequence of making me an expert on a topic at the fringes of regenerative biology where, with few exceptions, the animal models of choice were *Hydra*, planarians, zebrafish, salamanders, and newts. But there *were* exceptions relevant to my work, and three in particular would inform the project. The first was the mammalian digit tip model originally described by Richard Borgens (Borgens, 1982). Amputation of the digit tip (bone and soft tissue) in young adult mice induced a healing response that was sensitive to the level of amputation: an amputation distal to the nail bed induced a regenerative response whereas an amputation proximal to the nail bed stimulated fibrotic repair (Borgens, 1982; Fernando et al., 2011; Neufeld & Zhao, 1995). Amputations through more proximal elements never regenerated (Neufeld, 1980). One thing that struck me as odd though was that this model of regeneration was conserved across all those mammals in which it had been examined, including humans (Illingworth, 1974; Muneoka, Allan, Yang, Lee, & Han, 2008; Neufeld, 1980; Singer, Weckesser, Géraudie, Maier, & Singer, 1987). The second was the WIHN model of hair follicle neogenesis (rev. in Wang et al., 2015). The defining feature of this wound healing model are *de novo* generated hair follicles in the center of the wound separated from the wound margins by a ring of scar tissue. In addition to the lack of new hair follicles at the margins, regeneration was age-dependent

with very large wounds necessary in older animals to generate new hair follicles (Ito et al., 2007). Moreover, it was also apparent that the new hair follicles in this model were unpigmented and lacked arrector pili muscles (Billingham & Russell, 1956), a fact which suggested that this phenomenon represented incomplete regeneration. The third model was a complete hole punch through the ear pinna. As originally described in the rabbit ear, this model demonstrated the progressive regeneration of cartilage, hair and full-thickness skin (Vorontsova, 1960). Subsequent investigations in rabbits showed that regeneration was sex-dependent, with males more likely to close the open holes than females (Joseph & Dyson, 1966; Williams-Boyce & Daniel, 1980). Cross-species comparisons suggested that hole closure alone, without appropriate histological evaluation of the new tissue, could confound a conclusion that regeneration occurred (Williams-Boyce & Daniel, 1986). Together, these models convinced me I needed to include multiple healing models, provide side-by-side comparison with identical injuries in lab mice, include both sexes in my analysis, and where possible, use older animals.

With these ideas in mind, I laid out a plan and headed back to Kenya for a third summer. I could partially reject the notion that *Acomys* exhibited WIHN because the samples I already collected showed new follicles across the entire wound bed (Fig. 3). Moreover, I found new arrector pili muscles and sebaceous glands associated with the regenerated spiny hairs (awl hairs) and guard hairs. What I needed to add was a comparison of similar sized wounds in mice. Thus, I added the observation that when I made identical large wounds in very young Swiss Webster mice, the result was a scar with no new hair follicles in the wound bed. While completing these wound healing experiments, I focused my remaining attention on adding an ear punch assay to the dataset I had in hand. Work on the Murphy's Roth Large (MRL/+) strain of mice had suggested that complex tissue regeneration in mammals was simply a matter of genetics as these and closely related inbred strains could rapidly close small, 2mm ear holes (Clark, Clark, & Heber-Katz, 1998; Kench et al., 1999). Having recently come across a paper demonstrating that middle-aged C57BL/6 mice could close 2mm ear holes just fine (Reines, Cheng, & Matzinger, 2009), I became suspicious that something other than regeneration might be able to explain the closure of small holes. Based on this intuition, I chose to use a 4mm hole to test regenerative ability in *Acomys* and Swiss Webster mice. Using males and females to test for a sex bias, I punched the ears and waited. During the first 2 weeks the ear holes looked similar between the species and I resigned

myself to accept the possibility that this assay would fail to elicit a regenerative response from *Acomys*. What a difference a week can make. Between the second and third week a profound disparity emerged between the two species. Notably, new tissue began filling in the *Acomys* ear holes. Approximately 40 days after the initial punch, the spiny mouse ear holes had filled in with new tissue whereas in the Swiss mice, the holes remained open and unchanged lest for a bit of new scar tissue that had formed in the margins. Whatever excitement I experienced seeing new hair follicles emerge in skin wounds paled in comparison to what I witnessed for the first time that summer. I now understood that what I had uncovered was something truly remarkable.

After many months of careful preparation, I drafted a manuscript containing the data I had collected. The paper presented the skin tearing behavior, quantification of skin weakness, comparative wound healing in small and large wounds, and finally, the ear pinna regeneration experiments. As peer-review goes, it was a slog, although in retrospect the reviews were quite positive. They were rightfully skeptical and wanted to see a deeper molecular characterization of regeneration while also providing more “mechanistic data.” Luckily, the editor was keen on the manuscript and understood the limitations of working with animals in the field (i.e., I couldn’t FACS sort ear pinna cells or isolate and expand fibroblast lines for cross-species transplantation, etc.). And so again, with a modicum of luck, it was accepted, and I had the distinct pleasure of introducing the scientific community to a fascinating new mammalian regeneration model (Seifert, Kiama, et al., 2012).

2.4 The model matures: Understanding complex tissue regeneration in a mammal

If the animal model was going to mature to the point where I could do more controlled experiments, I needed to address tractability. While working with animals in Kenya would allow me to collect more extensive comparative data between *Acomys* and nonregenerative sympatric species, I needed to establish a laboratory colony in the States if I wanted to begin unpacking the cellular and molecular basis for enhanced regenerative ability in mammals. Again, luck would facilitate progress on this front. After poking around online, I discovered that spiny mice could sometimes be acquired in the pet trade. As it turned out, Ben Cole, a former technician in a neighboring lab managed a reptile shop in Gainesville called Pangea. He also helped care for a very large snake collection owned by Dr. Elliot Jacobson, a retired veterinary pathologist, who, as it turned out, just happened to have a modest

colony of pet *Acomys*. When I mentioned to Ben that I was looking for spiny mice he brought me with him to Elliot's house one evening. Elliot had inherited his animals in the early 1990s from the Phoenix Zoo and he was more than happy to give me some breeders, especially after I explained why I was trying to start a colony. In fact, Elliot eventually gave me all his animals, some 40 in total, and Malcolm and I began breeding them in our homes as pets. Ultimately, we would donate these pets to our respective Universities as research animals and they would eventually become the founders for my colony at the University of Kentucky and Malcolm's colony at the University of Florida.

When I started my own lab in Kentucky, spiny mice allowed me to focus on trying to understand the mechanistic basis for interspecific variation in regenerative ability. In attempting to derive solutions to this problem, spiny mice had suddenly provided a system in which to develop new hypotheses while also testing existing ones. Although the skin phenotype had initially propelled the project, it was the ear pinna injury model that excited me most. As a composite tissue, the ear pinna contains skin, adipose tissue, skeletal muscle, nerves and elastic cartilage (Gawriluk et al., 2016). This provides a means to examine the regenerative capacity for these tissues as an assemblage, similar to appendage regeneration in other vertebrate models, but also for each of these tissue types individually. Moreover, external ear pinnae are a mammalian synapomorphy (Manley, 2012), making them a perfect structure to assess regenerative ability across a wide range of species. The pinna is easy to access and because of the elastic cartilage core, ear holes exhibit little to no contraction during healing. Lastly, there are really only two basic mechanisms to repair the ear injury: regenerative healing or fibrotic repair. Almost no other regeneration model described thus far affords this type of comparison between closely related species other than the lack of tail regeneration in certain lizards (Dunoyer, Seifert, & Van Cleve, 2020) and the loss of head regenerative ability in certain planarian and annelid species (Bely & Sikes, 2010; Needham, 1952).

Naturally, Malcolm had also become fascinated with the *Acomys* model, and with his own animals in Florida he was keen to continue using them to explore regeneration. While I felt it was within my rights to simply take the entire system with me to establish my own lab, who was I to selfishly claim the system as my own. If Malcolm wanted to continue working with *Acomys*, then I thought we just needed to establish boundaries to avoid direct competition. I laid out my reasons for wanting to use the ear pinna and asked if he could leave my students and I to focus on this model moving forward.

In exchange, I would abstain from working on the skin for the foreseeable future. Anything else was fair game for either lab. With a gentleman's agreement in place a deal was struck.

As a student of the historical and contemporary regeneration literature, I relished the idea of developing a new system, and potentially a community, that might make an impact in our field. With a burgeoning *Acomys* colony in Kentucky, I established two parallel spiny projects. The first sought to explore the immune response to injury across healing types and species. Specifically, I wanted to test the hypothesis that loss of regenerative capacity in mammals resulted from a bias in the injury response toward adaptive immunity and a strong inflammatory response (Godwin & Brockes, 2006; King, Neff, & Mescher, 2012; Mescher & Neff, 2005). The second centered on activation of post-mitotic (differentiated) cells at the injury site and their ability to re-enter the cell cycle, proliferate and coordinate regeneration of the missing tissues. Although dedicated stem cell pools maintain mammalian tissues undergoing lifelong physiological turnover (e.g., intestine, blood, epidermis, hair follicles, etc.) (Levi & Morrison, 2008), there was no solid evidence that cells in differentiated mammalian tissues could undergo the type of dedifferentiation and proliferation commonly observed during epimorphic regeneration in salamanders, newts and zebrafish (Hay & Fischman, 1961; Lo, Allen, & Brockes, 1993; Oberpriller & Oberpriller, 1974; Poss, Wilson, & Keating, 2002; Sandoval-Guzman et al., 2014). Instead, mammalian cells were generally regarded as refractory to the triggers that stimulated cell proliferation in regenerating species (Pajcini, Corbel, Sage, Pomerantz, & Blau, 2010; Tanaka, Gann, Gates, & Brockes, 1997). With these two lines of inquiry as guideposts, my new students and I set ourselves to work.

First, using our 4 mm ear punch assay, we established the regenerative equivalency of our captive *A. cahirinus* with the wild-caught *A. percivali* and *A. kempfi* (Gawriluk et al., 2016; Seifert, Kiama, et al., 2012) and our characterization was independently reproduced by Gustavo Tiscornia's group at the University of Algarve (Matias et al., 2016). These studies also demonstrated that wild murids and laboratory mouse strains, including MRL/MpJ mice, could not close 4 mm ear punches, let alone regenerate functional tissue. In fact, our study uncovered that 2 mm holes were small enough to be filled with scar tissue which underscored the importance of coupling ear closure rates with cellular analysis when assessing regenerative *versus* fibrotic repair. An important feature of epimorphic regeneration observed in other vertebrate and invertebrate systems is that under normal

environmental conditions, and among healthy animals, regenerative healing always occurs in response to injury. Likewise, in nonregenerating species, regeneration never occurs. Here again, results in *Acomys* reflected other regeneration models and also revealed that males and females regenerated equally well (Gawriluk et al., 2016; Matias et al., 2016; Seifert, Kiama, et al., 2012).

Cellular and genetic comparison of tissue healing in *Acomys* and *Mus* established key aspects of the regenerative microenvironment that were consistent with epimorphic regeneration. First, whereas *Mus* exhibited few cycling cells and almost none that were mitotically active, *Acomys* activated a robust proliferative population that persisted during new tissue formation (Gawriluk et al., 2016; Seifert, Kiama, et al., 2012). Second, gene and protein data demonstrated that extracellular matrix production during fibrotic repair was dominated by type I collagen with lesser contributions from other wound-induced ECM molecules (e.g., *Tenascin-C*, *Mmps*, *Col3a1*, etc.). In contrast, the wound environment observed during regeneration in spiny mice was rich in Fibronectin and matrix remodeling enzymes (*Mmp9*, *Mmp13*) while also maintaining high levels of Tenascin-C. This demonstrated a regenerative ECM composition that was rich in molecules supporting proliferation and migration and including those that serve to resist collagen aggregation (Calve, Odelberg, & Simon, 2010; Calve & Simon, 2012; Monaghan et al., 2012; Satoh, Hirata, & Makanae, 2012; Satoh, Hirata, & Satou, 2011; Seifert, Monaghan, et al., 2012). When vertebrate appendages regenerate, newly generated epidermis covering the injury needs to expand as new tissue is generated beneath. This neoepidermis covering the injury is often referred to as the wound epidermis and similar to wound healing, it is required for regeneration to proceed. Here again, our RNA-seq and immunohistochemical data showed that a transitional epidermis formed early during healing in *Mus* and *Acomys*. Whereby the epidermis quickly differentiated into a stratified, squamous epithelium during fibrotic repair, it remained in a transitional state, devoid of the lamina densa (i.e., the lower portion of the basement membrane) until the ear hole closed in *Acomys*. Lastly, a key requirement for complex tissue regeneration is the presence of an intact nerve supply that facilitates blastema formation and supports cell proliferation (rev. in Sinigaglia & Averof, 2019; Stocum, 2011). During new tissue formation in the ear pinna, we observed the invasion of peripheral axons from the proximal margin of the ear, a finding later corroborated by Malcolm Maden (Maden, 2018) and we ultimately demonstrated that new axons had innervated the regenerated tissue

(Gawriluk et al., 2016). Thus, the collective observation of localized cell proliferation, a growth promoting ECM, formation and retention of a transitional epidermis and the regeneration of peripheral nerves, cemented *Acomys* as a bonafide vertebrate model of complex tissue regeneration.

Building on this work, a comparative analysis of regeneration and fibrotic repair using *Acomys* and other mammals has provided a deeper understanding of the salient features that contribute to the trajectory of healing in different mammals. Results from several labs examining *Acomys* and non-regenerative murids have detailed phenotypic differences within inflammatory and adaptive immune cell populations that have helped refine existing ideas about how these cells may influence regenerative healing (Brant, Yoon, Polvadore, Barbazuk, & Maden, 2016; Cyr et al., 2019; Gawriluk et al., 2020; Simkin, Gawriluk, Gensel, & Seifert, 2017). While studies in amphibians and fetal mammals have fed the idea that strong inflammatory and adaptive immune responses are antagonistic to regeneration (Godwin & Brockes, 2006; King et al., 2012; Mescher & Neff, 2005), data from *Acomys* suggests a more nuanced interpretation of the existing data. For example, a detailed examination of the inflammatory microenvironment over a 20 day time course in *Acomys* and *Mus* showed that while regeneration was associated with significantly reduced levels of two pro-inflammatory cytokines (IL-6 and CXCL1) compared to fibrotic repair, both types of healing elicited a common set of pro-inflammatory cytokines and chemokines (Gawriluk et al., 2020). A time course analysis of innate immune cells determined that neutrophils were significantly more abundant during fibrotic repair immediately after injury, but that macrophages arrived in equal numbers and persisted regardless of healing type (Simkin et al., 2017). Although macrophages are required for regeneration in salamanders and zebrafish (Godwin, Pinto, & Rosenthal, 2013; Petrie, Strand, Tsung-Yang, Rabinowitz, & Moon, 2014), a finding we extended to *Acomys* (Simkin et al., 2017), what remains less clear is how they specifically regulate regeneration. Recent work in our lab suggests that interspecific differences in macrophage subtypes can partly explain how macrophages direct local cell populations to differentially respond to injury (unpublished data). Comparative *in vitro* and *in vivo* experiments examining circulating and resident macrophages reveal that *Acomys* macrophages uniquely secrete several pro-angiogenic and antifibrotic factors that contribute to accelerated revascularization of the injury site and production of pro-regenerative matrix remodeling enzymes (unpublished data). Similarly, a phenotypic analysis of circulating and bone-marrow derived neutrophils from wild and captive *Acomys* and *Mus* revealed that *Acomys*

neutrophils are significantly more phagocytic and that humoral killing capacity is significantly stronger in spiny mice (Cyr et al., 2019). Together, these data point to subtle alterations in the innate immune response that maintain effective pathogen clearance and inflammatory resolution, while simultaneously promoting a regenerative microenvironment. It is also worth noting that our cursory analysis of T cell behavior during regenerative healing supports that *Acomys* exhibit a strong adaptive immune response to injury, possibly even stronger than the T-cell infiltration observed in laboratory and wild-caught murids (Gawriluk et al., 2020). Further characterization of immune cell phenotypes and manipulating specific macrophage populations in the context of regenerative healing and fibrotic repair is the natural evolution of where this work is headed.

Although phenotype switching among immune cells, and the secreted factors they supply, regulate the intensity and time course of inflammation (rev. in Eming, Hammerschmidt, Krieg, & Roers, 2009; Martin & Leibovich, 2005), inflammation remains a key feature of the general injury response. Fibrotic repair and regenerative healing seemingly represent two extremes of the final healing process, and yet the degree to which the innate immune response and inflammation promotes one healing type over the other remains unresolved. The ability to study how an identical injury results in two different healing types suggests three general models that can explain how the early responses to tissue damage might differentially affect healing outcomes. First, if the endpoints of fibrotic repair and regeneration lie along a smooth continuum, where local cells have the potential to heal by either means, then identical injuries could either trigger a response that is unique enough for each outcome that the two paths immediately diverge toward fibrosis or regeneration or instead, injury could trigger a response that is generally similar and instead differs primarily in its magnitude (Aloysius, Saxena and Seifert, 2021). Our current results revealing differences in immune cell phenotype points to the former and suggests that altering macrophage phenotypes during fibrotic repair could, in principle, shift the healing outcome. A third possibility is that fibrotic repair and regeneration represent two totally different responses that occur in a tissue-specific or species-specific manner and are relatively independent of the local innate immune response. Thus, upon the resolution of inflammation each healing outcome proceeds according to cell autonomous mechanisms regulated by local responding cell types (e.g., endothelial, keratinocyte, fibroblast, etc.) (Phan et al., 2020).

Support for distinct healing trajectories following inflammatory resolution comes from the observation that during epimorphic regeneration local

cells must proliferate to generate new tissue. Unlike regeneration in many invertebrate systems where pluripotent stem cells are present in adult animals, regeneration in vertebrates relies on lineage-restricted cells in differentiated tissues to act as progenitor cells for new tissue formation (rev. in Kragl et al., 2009; Monaghan & Maden, 2013). Hereto, *Acomys* is beginning to offer insight into how cell intrinsic properties among species may confer differences in regenerative ability. In an attempt to explore the proliferative capacity of connective tissue fibroblasts across regenerating (spiny mice and rabbits) and nonregenerating (mice and rats) mammals, we have uncovered a remarkable capacity for cells from regenerating species to resist oxidative stress (Saxena, Vekaria, Sullivan, & Seifert, 2019). Whereas cells from mice and rats activated a senescence program in response to oxidative stress, exogenous hydrogen peroxide was detoxified by a remarkably efficient glutathione redox reaction. Moreover, this resilience to oxidative stress extended to their mitochondria as well. Metabolic analysis of ear pinna fibroblasts suggested that differentiated cells from spiny mice and rabbits rely on aerobic glycolysis to meet their energy demands, a property usually observed in stem cells (Saxena et al., 2019). Our previous work also documented widespread activation of p21 and p27 throughout healing ear tissue in *Mus* suggesting that injury stimulates cellular senescence during fibrotic repair (Gawriluk et al., 2016). In contrast, we rarely observed upregulation of these tumor suppressor proteins in regenerating tissue, although we detected them in nearby differentiated tissue. Ongoing work in our lab has extended these findings *in vivo* and provides a mechanism that would seem to explain how injury stimulates cell cycle progression in cells that are poised to proliferate (unpublished data). Together with our immunological work, a picture is emerging for how cell autonomous and noncell autonomous factors have evolved in the *Acomys* lineage to facilitate complex tissue regeneration in response to injury.

2.5 Potential for regeneration in other tissue and organ systems

After leaving the University of Florida, Malcolm moved forward with a strategy to assess regenerative ability in the skin and other organ systems. Analyzing skin healing in *A. cahirinus*, his lab documented parity with our original work in *A. percivali* and *A. kempfi* (Brant, Lopez, Baker, Barbazuk, & Maden, 2015; Brant et al., 2016). Building on this work, Cheng Ming Choung's lab confirmed the regenerative ability of *A. cahirinus* skin and showed that all hair types are regenerated and that they are pigmented in contrast with

the WIHN mode of healing in C57BL/6 mice which only regenerate unpigmented zig-zag hairs (Jiang, Harn, Ou, Lei, & Chuong, 2019). Recent work from this same group showed that tissue stiffness across the wound bed appears to help regulate the natural regenerative properties of spiny mouse skin (Hans et al., 2021). These studies confirm the uniqueness of *Acomys* skin regeneration and in light of mouse studies which have pinpointed specific fibroblast phenotypes that confer enhanced regenerative ability (Driskell et al., 2013; Jiang et al., 2018; Rognoni et al., 2016) provide a platform for understanding the behavior of dermal fibroblasts across these systems. From the standpoint of utilizing this type of information to stimulate regeneration, multiple groups have recently demonstrated how enriching specific fibroblast subsets in skin wounds can be used to promote regeneration in adult mouse wounds (Mascharak et al., 2021; Phan et al., 2020). It will be important to determine if similar fibroblast subsets are enriched in *Acomys* through comparative scRNA-seq experiments or if they deploy a unique strategy to facilitate endogenous skin regeneration.

In addition to the skin, recent work in *Acomys* has also demonstrated a propensity of other organs to resist fibrotic damage. In response to injury, the kidney, heart and spinal cord appear to activate cytoprotective mechanisms which allow these animals to resist damaging fibrosis that would otherwise compromise organ function in a mouse or rat (Koopmans et al., 2021; Okamura et al., 2018; Peng et al., 2021; Streeter et al., 2020). The heart provides an intriguing example. Recent work from Malcolm's group suggested that in response to a myocardial infarct, the *Acomys* heart can recover a good deal of heart function as measured by echocardiography (Qi et al., 2021). More recently, a collaborative effort in Kentucky in parallel with a project at Hubrecht University led by Kerstin Bartscherer's group documented a remarkable resistance to heart injury. In response to a myocardial infarct, either by a permanent left anterior descending artery (LAD) ligation (Peng et al., 2021) or an ischemia-reperfusion infarct (Koopmans et al., 2021), our respective groups observed recovery from identical injuries which proved lethal to inbred and outbred mouse strains. In both studies, proliferation of endothelial cells and rapid angiogenesis were observed which contributed to a well-vascularized infarct area that allowed the heart to recover ventricular outflow. Our group found that a large fraction of *Acomys* cardiomyocytes exhibited a T-type calcium channel that is normally found in embryonic cardiomyocytes and the Koopman study was able to demonstrate an expansion of the local cardiomyocyte pool. Obviously, more work is needed to determine the cellular mechanisms at

play in this in other organs, but with an increasing battery of validated reagents for *Acomys* and a fully annotated genome, the horizon for these studies appears bright.



3. Reproduction and developmental biology

In the fields of observation chance favors only the prepared mind.

Louis Pasteur, from a lecture delivered at Universite de Lille, France (07.12.1854).

3.1 Prologue

While I (Peter) have been fortunate during my research career to have had a wealth of scientific experiences in the field of biology, I entered the field of spiny mouse reproduction rather late and by chance. I have always been interested in the intimate and fascinating process of reproductive biology, especially among the hairy, milk-producing mammals. Although my graduate project was initially to be on the reproductive biology of green turtles, that opportunity, also by chance, was withdrawn due to the unexpected disappearance of my designated PhD supervisor which happened 3 weeks before I was due to move from my alma mater, the University of Tasmania, to the Australian National University to start my PhD. I was then offered, as last-minute alternatives, two new projects from the Department of Zoology: a physiological study of a beautiful semi-aquatic reptile, the eastern water dragon (*Intellagama leseurii*), or a field study of that enigmatic, semi-aquatic icon of the Australian fauna, the platypus, *Ornithorhynchus anatinus*. While I was sure that the eastern water dragon would make for an interesting study species, I immediately knew that the platypus would be my choice, and I rapidly penned a letter to accept what would become the first PhD field study on the platypus. Four years of field work produced my PhD (Temple-Smith, 1973) which led to a 3-year Ford Foundation Post-doctoral Fellowship in the United States at Cornell University researching sperm maturation and fertilization in rabbits, hamsters, Australian brush-tailed possums (that I had brought with me from Australia) and Virginian opossums with Professor J. Michael Bedford, a Cambridge graduate in veterinary medicine who became an important mentor not only for my postdoctoral fellowship but the rest of my academic career. On my return to take up a position at Monash University I was able to extend my research interests into mammalian reproduction with projects on species from the three primary mammalian lineages: the monotremes (Temple-Smith & Grant, 2001), marsupials (Temple-Smith, 2003) and

eutherians (Verma, Holland, Temple-Smith, & Verma, 2012; Wang, Catt, Pangestu, & Temple-Smith, 2009), including human reproduction and IVF (Temple-Smith, Southwick, Yates, Trounson, & De Kretser, 1985).

Although I had been doing laboratory research on other rodent models, my exposure to spiny mouse biology happened through a fortuitous collaboration with a young and enthusiastic early career researcher, Hayley Dickinson, who had completed her PhD on renal function in the spiny mouse (*A. cahirinus*) (Dickinson, Moritz, Wintour, Walker, & Kett, 2007) and was expanding into pregnancy and developmental studies using these animals. At the time of our encounter, I had no idea about, nor direct interest in, the spiny mouse as a research model. However, this was about to change very rapidly during our joint supervision of a PhD student and the completely unexpected findings from her research that spiny mice menstruated (Bellofiore et al., 2017; Bellofiore & Evans, 2019).

Until very recently, there was scant interest in the spiny mouse as a model for studies of reproduction or developmental biology. Regarded as just another estrous-cycling murid rodent requiring special security features for captive colonies because of its status as a pest species in Australia, the costs of maintaining spiny mice did not justify the expenditure when compared with the advantages of other laboratory murids such as *Mus musculus*. And *Mus* has the advantage of a full battery of molecular reagents, transgenesis, biomarkers, genetic strains and disease models, where comparatively few exist for the spiny mouse model. Until recent discoveries showed the potential of the spiny mouse to provide certain more effective human-disease models, there has been little incentive to entice researchers working with mammals to consider adopting spiny mice into their studies.

The spiny mouse colony at Monash University was the brainchild of David Walker, a perceptive, enthusiastic and well-respected physiologist and neurobiologist with a passion for the comparative approach to study problems in biomedical research. Having understood the opportunities offered by the spiny mouse as a research model for developmental biology research, he started the colony with five founder pairs that he managed to import from the University of Amsterdam in 2002. The Amsterdam colony had been started from four pairs of founders acquired from the University of Geneva suggesting that the current Monash colony has extremely low genetic diversity from years of bottlenecking and inbreeding. As I have recently found in my short personal journey of discovery with the spiny mouse, establishing a spiny mouse colony in Australia was a portentous affair! David Walker's prescience in starting this colony changed the

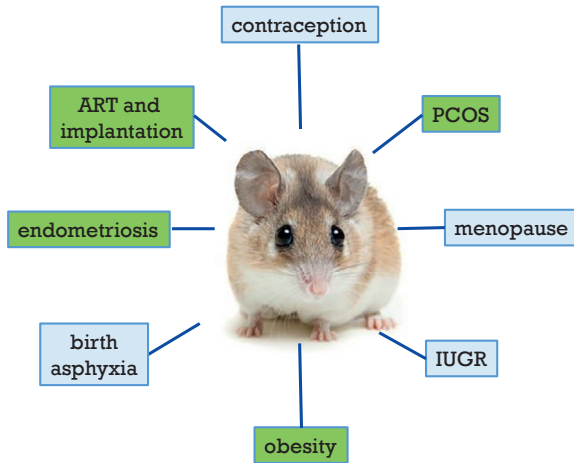


Fig. 4 Current (blue) and potential (green) spiny mouse models for women's health research. Assisted reproductive technologies (ART), intrauterine growth retardation (IUGR), polycystic ovary syndrome (PCOS).

direction of animal modeling in neurobiology and development while also contributing a new model for reproductive research, especially for several aspects of women's health (Fig. 4).

3.2 Menstruation and premenstrual tension: Pseudoserendipitous discoveries in female spiny mouse reproductive biology

The perception of spiny mice as an inferior model for women's health studies compared to laboratory mice and rats changed abruptly with the completely unexpected discovery that *A. cahirinus* females from our captive colony naturally menstruated in a very human-like fashion (Bellofiore, Cousins, Temple-Smith, Dickinson, & Evans, 2018). This observation set spiny mice apart, not only from *Mus* and *Rattus*, but from all other rodents thus far studied.

The story behind this startling discovery began with a suggestion to a bright young PhD student, Nadia Bellofiore, that she map out in more detail the estrous cycle of female spiny mice in the Monash University colony. Although a paper on the spiny mouse estrous cycle had been published in 1981 (Peitz, 1981), we thought characterization of the estrous cycle could be improved using more recently developed techniques. Although we didn't know it at the time, our seemingly innocuous suggestion to re-examine the spiny mouse estrous cycle would produce a totally

unpredicted outcome for our understanding of female reproduction in this nonmodel rodent. Importantly, the discovery would elevate this relatively unknown species into more global scientific prominence as a new, rapidly emerging animal model for studying many aspects of human reproduction, especially as they relate to women's health (Fig. 4), in a profoundly new way.

Why was the demonstration of a menstrual cycle in the spiny mouse such an unexpected discovery? In part, because the spiny mouse remains the only rodent in which a menstrual cycle has been rigorously documented out of 2277 described species from 481 genera and 33 families (Carleton & Musser, 2005; Honeycutt, 2009). Although the cycle duration was short compared to women, (average cycle length, *Acomys* = 9 days; human = 28 days), we also discovered that endometrial changes characterizing the human menstrual cycle were mirrored in the spiny mouse endometrium during menstruation (Fig. 5). This discovery arose from naturally skeptical researchers in our group following clues that many would have dismissed based on an accepted adherence to dogma. The dogma in this case was that all rodents have estrous cycles; an idea so entrenched in the field that the occurrence of a menstruating rodent was considered unlikely, if not impossible.

The pathway to this finding followed the classical steps involved in evidence-based research. One day, Nadia approached her supervisors (Hayley Dickenson and me), about blood contamination in the vaginal smears she had been taking from a group of spiny mice in her project. Not surprisingly, Nadia was worried that there was a mechanical problem with her technique. We suggested that she practice again her lavage technique using laboratory mice, which she did and was confident, from the lack of blood in her practice *Mus* samples, that her technical skill in vaginal lavage was now competent. When she returned to vaginal lavage in the female spiny mice, she reported again that she was finding blood in the samples. This was the light-bulb moment for us. Maybe the bleeding wasn't an artifact at all! Had she instead, unlikely as it might seem, collected evidence that female *A. cahirinus* might have a menstrual cycle rather than the standard rodent estrous cycle? If so, we needed to collect additional experimental evidence. We then discovered that it was not uncommon for the animal house staff managing our colony to find traces of blood on the floor of the cages. However, when this had happened in the past, they had simply reported the blood to their superiors (management and researchers) who gave permission to euthanize these females over concerns that aggression was causing internal injuries. While this decision may have been appropriate under these circumstances, on at least some occasions, it is possible that the animal technicians may have been inadvertently finding menstrual blood from overt bleeders!

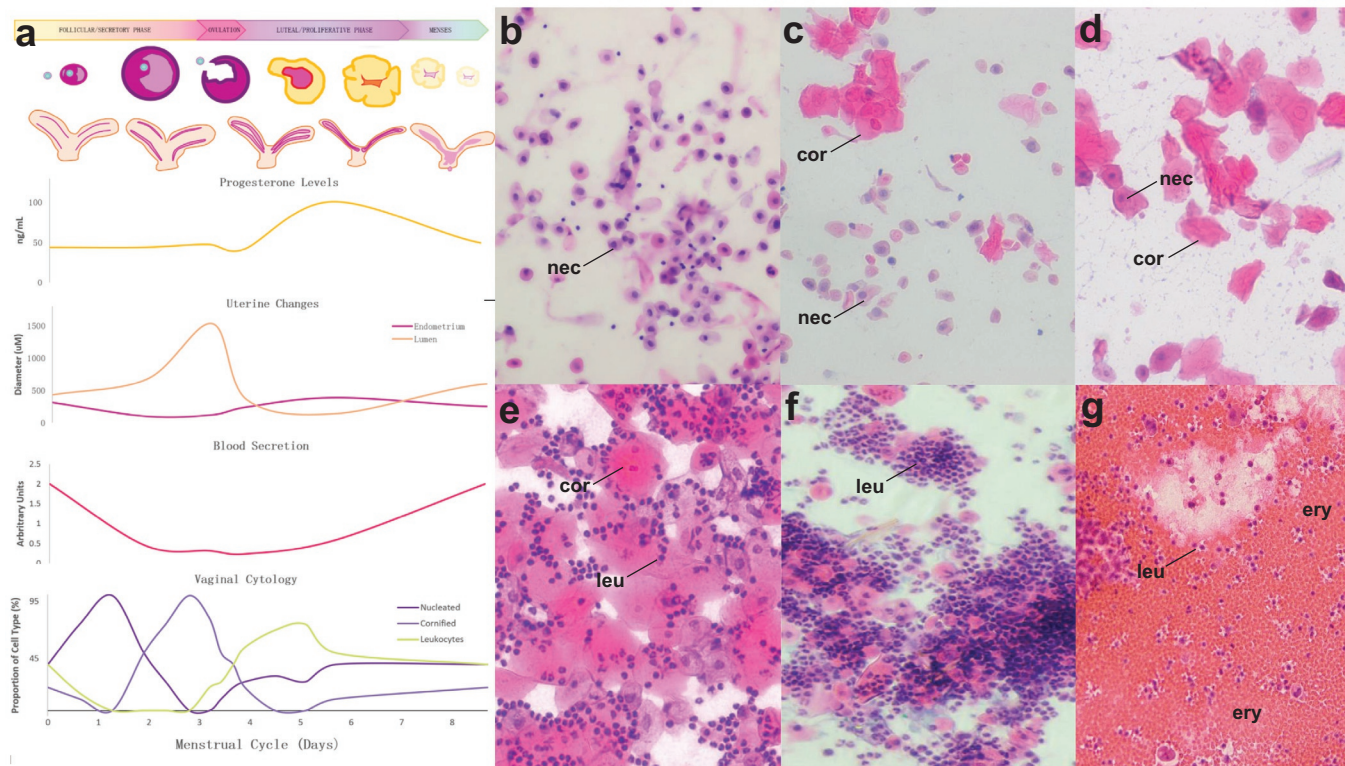


Fig. 5 See figure legend on next page.

Since captive colonies of spiny mice have existed in other parts of the world, it is interesting to contemplate why this discovery had not been made sooner. As the proverb suggests, discovery in science *does* favor the prepared. Given the infrequency with which menstruation has been observed in mammals, (menstruation is restricted to $\sim 2\%$ of all mammal species) perhaps it seemed preposterous to recognize it in a rodent. An alternative explanation is that our colony, due to its derivation from five founding pairs, had undergone some unlikely inbreeding-related mutation that resulted in the cyclical sloughing, rather than absorption, of the endometrial tissue at the end of each cycle. Perhaps this could explain the 1981 description by Peitz of spiny mouse estrous cycle? Prior to our description, menstruation had only been recorded in the higher primates (Emera, Romero, & Wagner, 2012), bats (Brasted, White, Kennedy, & Salamonsen, 2003; Rasweiler, 1991; Rasweiler & De Bonilla, 1992) and the elephant shrew (Brasted et al., 2003; Rasweiler, 1991; Rasweiler & De Bonilla, 1992; Van der Horst, 1946). Use of these species as laboratory models for studying menstruation is challenging. For the higher primates, the challenge is the cost of maintenance and growing ethical concerns about scientific research use. For the elephant shrew and bat species, it is access, captive facilities, specialist knowledge on their reproductive biology and the nonhuman style of their menstrual cycles that has hindered their use as experimental models.

Discovery of periods in the spiny mouse awakened significant interest in this species as a model for women's health. For example, in addition to menstruation, our lab through the work of Nadia Bellofiore has now demonstrated that spiny mice display the behavioral equivalent of a human-like premenstrual syndrome (PMS) during the late luteal phase (Bellofiore, Cousins, Temple-Smith, & Evans, 2019). PMS affects millions of women

Fig. 5 Spiny mouse menstrual cycle (A) summary of the changes in the ovary, reproductive tract, steroid hormones, menstrual flow and vaginal cytology during the spiny mouse menstrual cycle. B–G stage specific vaginal cytology: (B) early follicular stage (estrogen), duration ~ 2 days, cell type: nucleated epithelial cells. (C) Mid-follicular transition stage (estrogen), duration < 1 day, cell type: nucleated epithelial cells, cornified cells. (D) Late follicular stage/ovulation (estrogen), duration ~ 1.5 days, cell types: cornified cells. (E) Early luteal (estrogen, progesterone), duration < 1 day, cell types: cornified cells/leukocytes. (F) Late luteal (progesterone), duration ~ 2 days, cell types: leukocytes, some cornified cells. (G) Menstrual shedding, duration ~ 3 days, cell types: erythrocytes, leukocytes. Cor-cornified cell, ery-erythrocyte, leu-leukocyte, nec-nucleated epithelial cell. Adapted from Bellofiore, N., Ellery, S. J., Mamrot, J., Walker, D. W., Temple-Smith, P. and Dickinson, H. (2017). First evidence of a menstruating rodent: The spiny mouse (*Acomys cahirinus*). *American Journal of Obstetrics and Gynecology* 216, (40. e41-40. e11).

worldwide and is characterized to various degrees by symptoms such as nausea, abdominal pain and cramps, bloating, sore breasts, acne, irritability, food cravings and fatigue that may be severe enough to require some form of intervention. Because the mechanism behind PMS is still poorly understood, having a small animal model that displays symptoms may provide new information about the etiology of PMS, and possible treatments. The study design for this discovery was to monitor and score female spiny mouse behavioral responses during the menstrual cycle to restraint during vaginal lavage and to also track exploratory behavior, including female memory and social behavior and physiological changes using metabolic cages. The surprising findings were that, compared to controls, female spiny mice during premenstrual phases of the cycle were more reactive to constraint, moved less, spent more time in isolation in their communal cage, and increased their food consumption significantly compared to females in the late follicular phase of their cycle.

While menstruation has not yet been confirmed in female spiny mice in other captive colonies, and perhaps more importantly by spiny mice in the wild, it is intriguing to consider how and why this may have evolved. The frank release of blood by any species in the wild, from menstruation or otherwise, would be expected to have a highly negative selective value when compared with the standard rodent reabsorption of the engorged preimplantation endometrium during an estrous cycle. It seems counterintuitive that wild populations of spiny mice and other menstruating mammals would provide sensory cues from menstrual blood that enhanced the potential for predators to locate these important adult females. Selective removal of breeding females from a population by predators might be expected to have a negative selective pressure on population recruitment, especially in vulnerable or endangered species. However, in our colony only about 10–15% of females actually show overt bleeding from the vagina during menstruation; most females present with occult menstrual bleeding that can only be confirmed by vaginal lavage. We have observed a range of bleeding levels from light to heavy bleeders which suggests that this gradation in levels of bleeding depends on the extent of development of the endometrium during each cycle. This may be directly linked to different levels of circulating steroid hormones, in particular estrogens, progesterone and dehydroepiandrosterone (DHEA), which [Bellofiore and Evans \(2019\)](#) proposed is integral in spontaneous decidualization and menstruation in the spiny mouse. Assuming that this also occurs in wild populations of spiny mice, the selective pressure from predation would fall more directly on females that were

heavy bleeders and released blood from menstruation into the environment. In fact, for some predators with a keen sense of smell, even occult bleeding may be possible to sense by olfaction. For the spiny mouse to be a broadly accessible emerging model for women's health, we need to confirm that females in other captive colonies and perhaps also in wild *Acomys* populations also menstruate. There is already some evidence that females from the Kentucky colony menstruate (Ashley Seifert, Thomas Gawriluk, and Katarina Medger, personal communication). Clearly, an important future direction in spiny mouse menstrual research will be to include field studies of wild-caught populations to confirm, or otherwise, the existence of natural menstruation. These studies should also extend to other species in the Deomyinae which includes at least 19 species of *Acomys*, one species of *Deomys*, 33 species of *Lophuromys* and a single species of *Uranomys*, to determine if menstruation is a feature of other species in this group.

3.3 Menopause and reproductive senescence

Previous work in other labs and more recently by Evgenia George, a graduate student at Monash, has provided a detailed analysis of ovarian morphology and shown that, as in humans and a few other species, but unusual for rodents, the ovarian reserve develops prenatally with primordial germ cell/oocyte nests breaking down to form primordial follicles 2–3 days before birth. From a stereological analysis of an ontogenetic series of ovaries (prenatal to aged), Evgenia has shown that there is a significant age-related decline in the ovarian reserve from female spiny mice between 3 and 4 years of age (unpublished data).

More recently, work by Bellofiore and George have confirmed that this decline in ovarian reserve is part of another intriguing aspect of female spiny mouse reproductive biology: a human-like transition to menopause. No other rodent shows this pattern of gradual senescence, even those rodents that show an alternative form of senescence described as estropause (Battiston et al., 2017), which is accomplished rapidly at 9 to 15 months of age in estrous cycling murine rodents. The idea that spiny mice display a gradual transition to a menopause-like state with aging has revealed another fascinating characteristic of their reproductive biology. Recognition of this transition might have occurred earlier were it not for a self-imposed breeding criterion for our colony: that breeders are retired from breeding and culled after 12 consecutive litters. This meant that few females, or males, survived in the colony past 3 years of age. This was mostly a financial decision at

a time when the cost of maintaining the colony was significantly more than now. With the discovery of menstruation, and some financial relief on colony maintenance, we started to explore the reproductive lifespan of males and females in the colony, particularly with individuals between 3 and 4 years of age. This change has allowed us to collect new datasets on reproductive lifespan and to explore how aging affects reproductive capacity. In our captive colony we now know, from vaginal lavage and the histological presence of antral follicles, that many females up to 3 years of age retain the ability to cycle and breed but also that the degree of bleeding during menses in the 3–4 year old cohort declines significantly with a reducing ovarian reserve and gradual cessation of breeding. Recently completed plasma steroid hormone assays suggest that declining estrogen levels may affect the extent of endometrial proliferation in these older females resulting in lighter menstrual bleeds and even perhaps infertility. The gradual, rather than sudden, menopausal transition suggests that the spiny mouse is a more appropriate perimenopausal model than the current less natural rodent models for examining changing responses in the hypothalamic-pituitary-gonadal axis and other important pathways that influence this transition (Bellino & Wise, 2003; Zakaria et al., 2019).

Our menopause transition study highlights again, the ever-unfolding potential of the spiny mouse as an emerging model species for research in reproduction, development and aging. Our observations suggest that menopause in female spiny mice is similar to chimpanzees, where there is a gradual menopause-like transition with reproductively senescent females surviving the equivalent of 20% of their natural lifespan in reproductive senescence (Gould, Flint, & Graham, 1981). This highlights again the broad potential applications of the spiny mouse as an emerging model for various aspects of women's health (Fig. 4). It is currently the only small laboratory animal that provides a natural model for research into menopause.

3.4 Reproduction in male spiny mice

Compared to females, the reproductive biology of male spiny mice is poorly studied (Peitz, Foreman, & Schmitt, 1979). Despite the interesting human-like characteristics discovered in female spiny mice, the male reproductive system appears similar to most other rodents and there are no characteristics that have been discovered yet that suggest the male will provide any modeling that cannot already be achieved in the other more popular

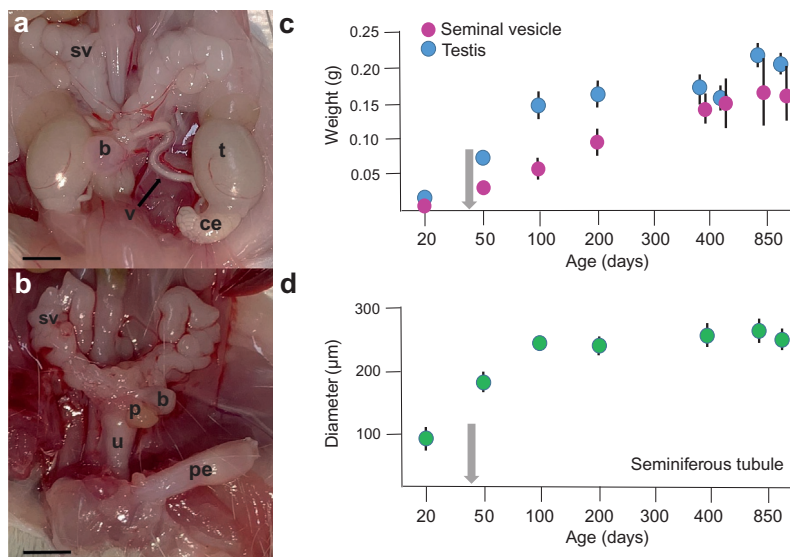


Fig. 6 Reproduction in the male spiny mouse. (A) *In situ* dissection of the male reproductive tract. Bladder (b), cauda epididymidis (ce), seminal vesicle (sv), testis (t), vas deferens (v). (B) *In situ* dissection of the male accessory glands of reproduction. Bladder (b), prostate (p), penis (pe), seminal vesicle (sv), urethra (u). (C) Changes in testis and seminal vesicle weights with age. (D) Changes in seminiferous tubule diameter with age. Panels (A) and (B): Photo credits, Peter Temple-Smith. Panels (C) and (D) adapted from Mamrot, J., Mamrot, S., Weybury, M., Wiadrowski, L., Temple-Smith, P., Dickinson, H. (2017) *Anatomy and physiology of the male reproductive tract in the spiny mouse (Acomys cahirinus) across the lifespan*. Poster, Annual joint meeting, Endocrine Society of Australia/Society of Reproductive Biology 2017, Perth, Australia.

laboratory rodents. However, there are some interesting reproductive features of male *Acomys* that deserve mention and warrant further investigation.

The male reproductive tract is typical of most other rodents with paired scrotal testes, epididymides and vasa deferentia (Fig. 6A and B). The inguinal canal is open and the testes and epididymides are retained in the scrotal sac by an inguinal fat pad as in laboratory mice and rats. The seminal vesicles are large and paired with ducts that join the distal ends of the vasa deferentia. Here they form paired ejaculatory ducts that pass through the prostate to drain into the prostatic urethra and paired bulbourethral glands that drain into the membranous section of the urethra before it passes into the penis to form the penile urethra. Although the penis contains an os penis, or baculum, which is not uncommon among rodents, preputial glands are lacking, a conspicuous absence among the rodents (Peitz et al., 1979). Peitz and

colleagues provide a detailed analysis of the male reproductive system where they reported testicular asymmetry in adult males, a fact we confirmed in the Monash colony, with the left testis consistently heavier than the right. We also confirmed that asymmetry in testis size is reflected in a similar asymmetry in daily sperm production. This arrangement may relate to increased spermatogenic activity associated with increased seminiferous tubule length and an increased number of Sertoli cells in the left testis. This study also reported that spermatogenesis commences before post-natal week 7 with spermatozoa first observed in the seminiferous tubules between 7 and 8 weeks (Fig. 6C and D, gray arrow). Others have reported that sperm production occurs between weeks 5 and 6 and that males can impregnate females by week 7 (Haughton et al., 2015). Although there are no detailed published studies on spiny mouse sperm morphology, general observations on sperm structure and motility suggest there that they are typically murid, with single hooked sperm head and rodent sized midpiece and principal piece (Breed, 1995). The epididymis appears normal in shape and from brief observations spermatozoa undergo a standard process of maturation during passage through the epididymis with acquisition of the capacity for motility in the corpus region and storage in the caudal region.

Observations on mating behavior in spiny mice, including a recent video-based study on Monash colony pairs (Mamrot, 2019), have reported various interesting and sometimes conflicting outcomes. Mating usually occurs within a day of parturition associated with post-partum ovulation and receptivity and mate selection appears to be female-driven with receptive females controlling access, mounting and withdrawal behavior. Multiple intromissions are usually observed with an absence of penile thrusting by the male and ejaculation signaled by a brief and unusual hind paw “wave” and an almost imperceptible penile locking in female tract with a duration of only a few seconds. In many rodents, including laboratory mice, successful ejaculation is confirmed by the presence of a copulatory plug which is derived from the coagulation of seminal fluids from the seminal vesicles, coagulating glands and prostate. Peitz et al. (1979) conducted a detailed study of accessory gland function to examine the mechanism behind coagulation. Using cross-species coagulation testing of seminal vesicle and coagulating gland secretions they showed reciprocal cross-reactivity between seminal vesicle and coagulating gland secretions from the spiny mouse and laboratory rat. They also showed that, while coagulating gland vesiculase from a broad range of rodents were able to coagulate seminal vesicle secretions from spiny mice, the spiny mouse coagulation gland secretions were

significantly less active on seminal vesicle secretions from a range of other rodents including the more closely related gerbil.

A controversial aspect of male spiny mouse reproduction concerns the presence or absence of copulatory plugs. Unlike in many rodent species, with the exception of one spiny mouse study in which plugs were commonly observed (Dewsbury & Hodges, 1987), most studies have concluded that no visible copulatory plug is formed in the vagina from semen coagulation following successful spiny mouse matings (Bellofiore, Ellery, Temple-Smith, & Evans, 2020; Dickinson & Walker, 2007; Peitz, 1981). In my coauthor's lab, an extensive study of female spiny mice immediately post-copulation found no evidence of plugs and a recent study in our lab confirmed these observations. The observations in our lab study by Jarrod McKenna were associated with vaginal lavage in an attempt to confirm at least the presence of large numbers of spermatozoa in the vagina post-coitus but, with the exception of a few spermatozoa in the lavages, there was no other evidence that ejaculation had occurred. Interestingly, despite the presence of few spermatozoa in the lavages, many of the females were subsequently found to be pregnant after the pair had been separated. This raises the possibility that during a brief penile lock, the ejaculation is intracervical or intrauterine in spiny mice—a possibility for future testing.

3.5 Precocial development

Compared with most other rodents, spiny mice exhibit extended fetal development *in utero* and are relatively mature at birth (Brunjes, 1990). Compared with altricial neonates (Fig. 7A and B) newborn spiny mice are able to see and hear, regulate their body temperature, move independently with relatively well coordinated movements, and forage for themselves within a few days of birth (Haughton et al., 2015). Whereas altricial rodents naturally ovulate large numbers of oocytes and give birth to an average of eight neonates following a gestation of 16–17 days, spiny mice usually ovulate a maximum of 3–5 oocytes, and often fewer. In the Monash and Kentucky colonies, births occur with an average of three neonates after ~39 days of gestation.

The rarity of precocial development among murids presents an opportunity to investigate and compare organ development with altricial laboratory rodents in which maturation of organ systems occurs postnatally. For instance, spiny mice complete lung and liver development (Oosterhuis, Mooren, Charles, & Lamers, 1984), and the majority of neurogenesis



Fig. 7 Comparison of development at birth. (A) Precocious spiny mouse pups (*Acomys* sp.). (B) Altricial laboratory mouse pups, (*Mus musculus*). Credit: <https://www.spectrumnews.org/news/toolbox/spiny-mice-model-risk-of-autism-from-maternal-infection/>

in utero (Dickinson, Walker, Cullen-McEwen, Wintour, & Moritz, 2005). Studies have also shown that, unlike altricial species, spiny mice responded later in gestation than altricial rats and gerbils to acute umbilical cord compression suggesting that this promotes fetal survival *in utero* (Robinson & Smotherman, 1992). Precocial development has also provided a unique opportunity for researchers to examine many aspects of newborn spiny mouse behavior. Using a suite of behavioral tests, studies have demonstrated motor activity, fine coordination, and social behaviors 1 day after birth (Ratnayake, Quinn, Daruwalla, Dickinson, & Walker, 2014). Moreover, motor coordination performance and an assessment of exploratory behavior measured 10–25 days after birth in spiny mice were found equivalent to these behaviors in adult mice (Ratnayake et al., 2014). Spiny mice have also provided a new experimental model to study the effects of altered gravitational conditions on fetal development (Sekulić, Božić, Božić, Borota, &

Ćulić, 2006). These studies make the spiny mouse an ideal subject for studies of both early ontogenesis and the mechanisms which result in species differences and highlights the spiny mouse as an emerging model for developmental biology (Brunjes, 1989).

3.6 Manipulating spiny mouse reproduction

In recent years, advances in our knowledge of spiny mouse biology have directed some research initiatives toward manipulation of the reproductive system to build new models to explore different facets of reproduction in spiny mice. Researchers in our department and linked institutes at Monash University have actively pursued some of the unanswered or controversial questions of reproduction in our colony, sometimes with interesting and often unexpected results. In this section of the chapter, I share four of these interesting experiences and outcomes and discuss how these experiments are influencing the next phases of reproductive research in spiny mice.

A robust and reliable superovulation protocol is a critical prerequisite for developing a successful *in vitro* fertilization program. Pasco et al. (2012) developed and optimized a technique resulting in a 100% success rate using this optimized protocol with an average recovery of 15×1 or 2 cell embryos per treatment. Considering that the average ovulation rate for unstimulated females is about 3 MII oocytes, this fivefold increase in ovulation/embryo production rate was highly significant and with appropriately timed collections would likely facilitate the development of IVF and an embryo culture system. The conundrum from this successful project was that embryonic development in the spiny mouse (Fig. 8A and B) is blocked at the 4-cell stage, not unlike the 2-cell block that originally delayed *in vitro* studies of mouse embryo development (Biggers, 2004). Jared Mamrot explored the development of cleavage-staged embryos in spiny mice and developed an important timed embryo cleavage series for comparison with the mouse and human (Fig. 8A and B). Despite a large number of treatments, he has not been successful in progressing spiny mice embryos past the 4-cell stage *in vitro*, although later stage cleavage embryos recovered from the oviducts are able to continue to the blastocyst stage given an appropriate *in vitro* environment (Mamrot, 2019). Recently, we tried unsuccessfully to repeat the (Pasco, Gardner, & Walker, 2012) superovulation protocol and have proposed the use of a human-based superovulation protocol (McKenna, Bellofiore, Dimitriadis, & Temple-Smith, 2021) as an alternative method that is currently being tested.

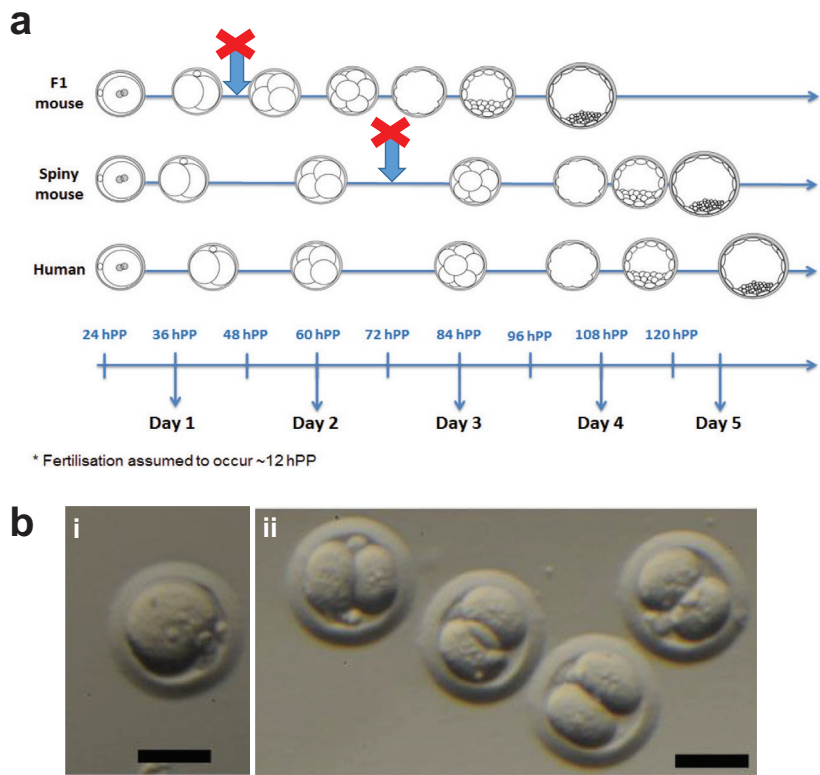


Fig. 8 Spiny mouse embryology. (A) Comparison of the time taken to reach the equivalent preimplantation cleavage stages in the F1 laboratory mouse, spiny mouse and human showing the similarity in timing between the spiny mouse and human. Note: *in vitro* cleavage blocks occur at the 2-cell stage in the mouse and the 4-cell stage in the spiny mouse. (B) Timed cleavage stage development in spiny mouse embryos from fertilization to expanded blastocyst. This figure is borrowed with permission from Mamrot, J. P. (2019). *Early developmental biology of the spiny mouse (Acomys cahirinus)*. Monash University.

In the second example, we were interested to determine if female spiny mice in the same cages are able to synchronize their menstrual cycles, similar to the way lab mice synchronize their estrous cycles. Synchronization of cycles is an important behavior in animal models used for reproductive studies because it simplifies time required to complete these studies. We were also interested to know if vaginal and other forms of stimulation elicit pseudopregnancy in spiny mice. To our surprise neither cycle synchronization nor pseudopregnancy could be induced in the female spiny mouse despite the use of various procedures including a daily vaginal lavage control,

progesterone injection, mechanical stimulation of the cervix and sterile mating for pseudopregnancy, and by exposing female groups to male-soiled bedding to assess synchronization *via* the Whitten effect (Bellofiore et al., 2020). Failure of the spiny mouse to show pseudopregnancy or cycle synchronization is an indication that the spiny mouse is not a typical rodent and more aligned in its reproductive strategy with higher-order primates.

Spiny mice from our colony have provided an interesting and informative model for studying the effects of birth asphyxia. The initial idea, developed by David Walker and his neurobiology group, used the precocious qualities, especially the long gestation, of the spiny mouse as an important component of the model to examine neural deficits after hypoxia at birth. The model compared the effects of a caesarian delivery at 37 days gestation, 2 days before normal term, of pups that were delivered without an asphyxia event compared with those that were given 7 min of asphyxia. This asphyxia event caused significant impairment of memory and learning tasks and caused various pathologies in the central nervous system including an inflammatory response and increased cell death compared to the controls (Hutton, Ratnayake, Shields, & Walker, 2009). The success of this asphyxia model began a series of related studies to explore the many possible applications of this model including demonstration that an acute pre-term hypoxic event caused mild neurological effects in various behavioral tests (Ireland, Dickinson, Fleiss, Hutton, & Walker, 2010) and that creatine supplementation protected various organ systems from the immediate pre-term effects of hypoxia-induced damage (Ellery et al., 2016; Ireland, Castillo-Melendez, Dickinson, Snow, & Walker, 2011). Stacey Ellery, an early career researcher at the Hudson Institute of Medical Research in Melbourne, more recently extended these studies on birth asphyxia and creatine supplementation in the spiny mouse model and is now translating findings on the protective nature of creatine supplementation into clinical applications; perhaps, the first time that biomedical science discoveries using the spiny mouse model have been translated into clinical studies (De Guingand, Ellery, Davies-Tuck, & Dickinson, 2019; Ellery et al., 2019, 2020).

Lastly, a recent study in our lab by Jarrod McKenna developed a sperm cryopreservation protocol for spiny mouse spermatozoa using a raffinose-based cryoprotectant with effective freezing and warming qualities that will enable us to preserve semen samples for colony insurance purposes and future development of artificial breeding techniques (McKenna, Catt, Pangestu, & Temple-Smith, 2020). Together, these advances have positioned the growing spiny mouse research community to develop transgenic

spiny mice, a feat that would allow this species to join a small and select group of mammals in which transgenesis has been achieved.



4. Future perspectives

4.1 Community

Whereas discovering a new phenomenon can be exhilarating, developing a new model can be arduous. Cultivating a new system can breed isolation or community. For every genuine inquiry of interest from the intellectually curious, there are the adulators, hungry for the possibility of something flashy and new that can be leveraged for another grant. For every thoughtful conversation about the challenges of working with a new model, there are those that only need your resources. For every believer, there is the naysayer. Luckily, our overall experience has been the collective former.

Although spiny mouse research colonies have existed almost continuously since the 1960s, research with these rodents has tended to remain in isolated pockets. The longest running colony (to our knowledge) is maintained at Tel Aviv University in Israel by Noga Kronfeld-Schorr whose broad biological interests have served to maintain a steady output of spiny mouse research over the years. The Tel Aviv colony is also close to wild spiny mouse populations allowing for continued maintenance of genetic diversity. Another longstanding colony remains at the University of Geneva, from which the Monash colony was derived, and is maintained by Michel Milinkovitch and Athanasia Tzika who have used these animals to explore questions in evolutionary and developmental biology ([Montandon, Tzika, Martins, Chopard, & Milinkovitch, 2014](#)). The Monash colony is approaching its 20-year anniversary and while the University of Kentucky colony was only started in 2013, its provenance dates back to the 1990s. Detailed publications on husbandry have been published and as detailed below, key molecular resources have been or are now under development. In addition to existing colonies in Europe, we have helped to establish new colonies at Caltech, Cornell, Liberty University, Emory University, University of Southern California, Hubrecht University, KAIST (Korea), Yale, Harvard and University of British Columbia. Gustavo Tiscornia also maintains a colony with several species of spiny mice at the University of Algarve.

After nearly a decade since the original publication of enhanced regenerative ability the response to *Acomys* as a useful model by other laboratories studying regeneration has been predictably mixed; an outcome that parallels

the reductionist thinking that has canalized the broader field of developmental biology. With the rise of genetic model organisms and a shift toward genetic description, research laboratories are loath to adopt a high-cost model that does not provide the wealth of molecular tools and transgenic technology present in a handful of model organisms. However, recent developments suggest this trend is slowly changing given an expansion of molecular resources and a steady stream of research that has begun to unpack the mechanistic basis for enhanced regenerative ability in these animals (Gaire et al., 2021; Simkin & Seifert, 2018). In contrast, since the discovery of menstruation, a premenstrual syndrome and a menopause-like transition in spiny mice, the reproductive biology community has been predictably more excited. This is not surprising given the absence of a small animal model of menstruation and the potential for the spiny mouse model to propel the field forward.

4.2 Molecular resources

Although the breadth of the standard molecular toolkit available for the laboratory mouse is not available for spiny mice, rapid technological advances should help to close the gap. A reference transcriptome representing 23 tissues (including heart, muscle, lung, brain, placenta, and gonads from an adult male and female) has been published (Mamrot et al., 2017) and more targeted *de novo* transcriptomes for skin healing (Brant et al., 2015) and complex tissue regeneration (Gawriluk et al., 2016) exist. These have been leveraged by other groups for comparative work exploring the evolution of regeneration (Wang et al., 2020) and have been important resources for our genome sequencing project. Through a joint effort including the Earlham Institute, EMBL-EBI, and the Sanger Institute, University of Kentucky and the University of Tel-Aviv, chromosome level assemblies for four *Acomys* species have been produced using a combination of PacBio long reads and Illumina 10 × reads (Biswas S., Wright D. et al. *in preparation*). While a rough genome for *A. cahirinus* was published in 2019 by the Broad Institute with tissue from animals at the University of Kentucky, investment in that sequencing project was limited and lacked the computational support to produce a draft genome useful to the growing community of *Acomys* researchers. The frozen *A. cahirinus* assembly produced in 2019 was shared broadly among researchers in the *Acomys* community in keeping with our commitment to open science among our burgeoning community. This chromosome-level assembly was annotated

by the Earlham Institute by integrating RNA-seq information and existing annotation from reference species (mouse, rat and human) and has provided a platform for RNA-seq, scRNA-seq and other comparative projects that rely on genomic information.

Very few spiny mouse primary cell lines have been developed and because of the dearth of colonies globally, few if any are readily available commercially. Both of our labs have developed a series of primary fibroblast cell lines that are useful for an array of studies. These also serve as an insurance strategy such that in the event that the Monash or Kentucky colony suffers a catastrophic event resulting in the complete loss of the colonies, we have the possibility with new and developing pluripotency and gamete generation methods to potentially re-create our colonies. We have also supported and assisted the development of colonies as outlined above, and a second colony of spiny mice was created for David Walker's group which is now located at the Royal Melbourne Institute of Technology. This colony used founders from the Monash colony to establish this new spiny mouse colony in Melbourne to provide us both with continuing active research, and also joint insurance colonies.

There is also now a growing incentive to establish other primary cell lines, such as embryonic stem cells, induced pluripotent stem cells, endometrial and other tissue-specific stem cells, fetal fibroblasts for *Acomys*-specific feeder layers and, in the more distant future, immortalized spiny mouse cell lines to provide what we anticipate will be an expanding need for all of these in future projects. A transgenic project is now underway at the University of Kentucky using the traditional ESC line approach which when coupled with the reproductive advances at Monash increase our chances for successfully producing the first set of transgenic spiny mice.

We have also found, often to our detriment, that our research on various aspects of reproduction in the spiny mouse is hampered by the dearth of experimental tools that are standard and commercially available for researchers working on other laboratory rodents, especially the mouse. Although the spiny mouse genome has now been sequenced and we have published a transcriptome for the species in our colony, compared with the vast resources available for laboratory mouse research, there is little in the way of genetic studies, specific biomarkers or antibodies for immunocytochemistry studies, and few bioassays available that are informative for the spiny mouse. Developing these tools will greatly improve our understanding of the various processes in reproduction and developmental biology that uniquely occur in the spiny mouse. For example, our data on menstruation and menopause

would be more informative, and even more interesting, if we had access to validated and optimized bioassays to measure with accuracy some of the important protein and peptide hormones of reproduction such as gonadotrophin-releasing hormone (GnRH), follicle stimulating hormone (FSH), luteinizing hormone (LH), inhibins and activins, other TGF-beta cytokines and anti-Mullerian hormone (AMH) in the spiny mouse. In our journey so far, it has been interesting to discover that a few commercial bioassays do work in the spiny mouse and even more interesting to find that these are often human, rather than rodent, assays. In addition to these needs, the future expansion of spiny mouse reproductive research is also likely to benefit from the development of organoid cultures and the isolation and characterization of organ-specific exosomes.

Acknowledgments

We would like to acknowledge those researchers that have pursued work with spiny mice in the past, often at great challenge. We would also like to thank the growing community of spiny mouse researchers that are committed to open science and that generously share information with each other about ongoing and future projects that facilitate our general understanding about these extraordinary rodents.

Beyond those named in this chapter, Dr. Seifert would like to acknowledge the tireless work of several Kenyan field assistants, namely, James Ekiru Erkanan who was instrumental in trapping and site scouting in the early development of the regenerative biology work. Also, Stanley Marete in the Department of Veterinary Anatomy and Physiology at the University of Nairobi who has maintained ongoing colonies of *A. kempi* and *A. percivali* and the staff and scientists at Mpala Research Centre. Greg Schultz for his mentorship in all things wound healing. Vanessa Ezenwa has been an amazing collaborator on the ecoimmunological aspects of spiny mice and continues to participate as part of our community. Thomas Gawriluk worked tirelessly in the field and back in the States to advance the spiny mouse model at the University of Kentucky. He also conducted the first reproductive biology studies in the Kentucky colony. Lastly, spiny mouse work in Dr. Seifert's lab has been generously supported by grants from the National Science Foundation (NSF) and the Office for International Science and Engineering (OISE) (IOS-1353713), the National Institutes of Arthritis and Musculoskeletal and Skin Diseases (NIAMS) (R01 AR070313), the National Institute of Dental and Craniofacial Research (NIDCR) (R21 DE028070) and a Grants-in-Aid of Research from the National Geographic Society.

Dr. Temple-Smith acknowledges David Walker, Stacey Ellery and the many researchers and students who, through their interest, enthusiasm and long hours in their laboratories, have provided the solid foundation on reproduction and development of spiny mice in the Monash colony that forms the basis of this chapter. Special thanks to Haley Dickinson for the productive collaborations on spiny mouse reproduction during her time at the Hudson Institute of Medical Research; Jared Mamrot for advancing our knowledge on embryo development in the spiny mouse and for permission to use (Fig. 8A and B) from his PhD thesis; Nadia Bellofiore for the fascinating discoveries, during her graduate research, of menstruation, premenstrual tension syndrome and absence of pseudo-pregnancy in the spiny

mouse; Evgenia George for her detailed research findings on the spiny mouse ovarian reserve and who, with Nadia, demonstrated the spiny mouse transition-to-menopause; Jarrod McKenna for his detailed and important work on sperm cryopreservation, mating behavior and post-partum reproduction. I also acknowledge the generous support from the Bill and Melinda Gates Foundation (OPP1171535—The Spiny Mouse for Screening Contraceptive), the Pharmaceuticals Research and Development branch at Bayer AG Germany, the Monash University Education Program in Reproduction and Development and the Monash Department of Obstetrics and Gynecology.

References

- Alhajeri, B. H., Hunt, O. J., & Steppan, S. J. (2015). Molecular systematics of gerbils and deomyines (Rodentia: Gerbillinae, Deomyinae) and a test of desert adaptation in the tympanic bulla. *Journal of Zoological Systematics and Evolutionary Research*, 53, 312–330.
- Aloysius, A., Saxena, S., & Seifert, A. W. (2021). Metabolic regulation of innate immune cell phenotypes during wound repair and regeneration. *Current Opinion in Immunology*, 68, 72–82.
- Battiston, F. G., Dos Santos, C., Barbosa, A. M., Sehnem, S., Leonel, E. C. R., Taboga, S. R., et al. (2017). Glucose homeostasis in rats treated with 4-vinylcyclohexene diepoxide is not worsened by dexamethasone treatment. *The Journal of Steroid Biochemistry and Molecular Biology*, 165, 170–181.
- Bauer, A. M., Russell, A. P., & Shadwick, K. R. E. (1989). Mechanical properties and morphological correlates of fragile skin in gekkonid lizards. *Journal of Experimental Biology*, 145, 79–102.
- Bellino, F. L., & Wise, P. M. (2003). Nonhuman primate models of menopause workshop. *Biology of Reproduction*, 68, 10–18.
- Bellofiore, N., Cousins, F., Temple-Smith, P., Dickinson, H., & Evans, J. (2018). A missing piece: The spiny mouse and the puzzle of menstruating species. *Journal of Molecular Endocrinology*, 61, R25–R41.
- Bellofiore, N., Cousins, F., Temple-Smith, P., & Evans, J. (2019). Altered exploratory behaviour and increased food intake in the spiny mouse before menstruation: A unique pre-clinical model for examining premenstrual syndrome. *Human Reproduction*, 34, 308–322.
- Bellofiore, N., Ellery, S. J., Mamrot, J., Walker, D. W., Temple-Smith, P., & Dickinson, H. (2017). First evidence of a menstruating rodent: The spiny mouse (*Acomys cahirinus*). *American Journal of Obstetrics and Gynecology*, 216 (40. e41–40. e11).
- Bellofiore, N., Ellery, S. J., Temple-Smith, P., & Evans, J. (2020). Pseudopregnancy and reproductive cycle synchronisation cannot be induced using conventional methods in the spiny mouse (*Acomys cahirinus*). *Reproduction, Fertility and Development*, 32, 363–372.
- Bellofiore, N., & Evans, J. (2019). Monkeys, mice and menses: The bloody anomaly of the spiny mouse. *Journal of Assisted Reproduction and Genetics*, 36, 811–817.
- Bellofiore, N., Rana, S., Dickinson, H., Temple-Smith, P., & Evans, J. (2018). Characterization of human-like menstruation in the spiny mouse: Comparative studies with the human and induced mouse model. *Human Reproduction*, 33, 1715–1726.
- Bely, A. E., & Sikes, J. M. (2010). Latent regeneration abilities persist following recent evolutionary loss in asexual annelids. *Proceedings of the National Academy of Sciences*, 107, 1464–1469.
- Biggers, J. D. (2004). Reflections on the culture of the preimplantation embryo. *International Journal of Developmental Biology*, 42, 879–884.
- Billingham, R., & Russell, P. (1956). Incomplete wound contracture and the phenomenon of hair neogenesis in rabbits' skin. *Nature*, 177, 791–792.

- Borgens, R. B. (1982). Mice regrow the tips of their foretoes. *Science*, 217, 747–750.
- Brant, J. O., Lopez, M.-C., Baker, H. V., Barbazuk, W. B., & Maden, M. (2015). A comparative analysis of gene expression profiles during skin regeneration in *Mus* and *Acomys*. *PLoS One*, 10, e0142931.
- Brant, J. O., Yoon, J. H., Polvadore, T., Barbazuk, W. B., & Maden, M. (2016). Cellular events during scar-free skin regeneration in the spiny mouse, *Acomys*. *Wound Repair and Regeneration*, 24, 75–88.
- Brasted, M., White, C., Kennedy, T., & Salamonsen, L. (2003). Mimicking the events of menstruation in the murine uterus. *Biology of Reproduction*, 69, 1273–1280.
- Breed, W. (1995). Spermatozoa of murid rodents from Africa: Morphological diversity and evolutionary trends. *Journal of Zoology*, 237, 625–651.
- Brunjes, P. (1989). A comparative study of prenatal development in the olfactory bulb, neocortex and hippocampal region of the precocial mouse *Acomys cahirinus* and rat. *Developmental Brain Research*, 49, 7–25.
- Brunjes, P. C. (1990). The precocial mouse, *Acomys cahirinus*. *Psychobiology*, 18, 339–350.
- Calve, S., Odelberg, S. J., & Simon, H. G. (2010). A transitional extracellular matrix instructs cell behavior during muscle regeneration. *Developmental Biology*, 344, 259–271.
- Calve, S., & Simon, H.-G. (2012). Biochemical and mechanical environment cooperatively regulate skeletal muscle regeneration. *The FASEB Journal*, 26, 2538–2545.
- Carleton, M. D., & Musser, G. G. (2005). Order rodentia. *Mammal Species of the World: A Taxonomic and Geographic Reference*, 2, 745–2142.
- Clark, R. A. F. (1996). *The molecular and cellular biology of wound repair*. Springer.
- Clark, L. D., Clark, R. K., & Heber-Katz, E. (1998). A new murine model for mammalian wound repair and regeneration. *Clinical Immunology and Immunopathology*, 88, 35–45.
- Cyr, J. L., Gawriluk, T. R., Kimani, J. M., Rada, B., Watford, W. T., Kiama, S. G., et al. (2019). Regeneration-competent and-incompetent murids differ in neutrophil quantity and function. *Integrative and Comparative Biology*, 59, 1138–1149.
- De Guingand, D. L., Ellery, S. J., Davies-Tuck, M. L., & Dickinson, H. (2019). Creatine and pregnancy outcomes, a prospective cohort study in low-risk pregnant women: Study protocol. *BMJ Open*, 9, e026756.
- Dewsbury, D. A., & Hodges, A. W. (1987). Copulatory behavior and related phenomena in spiny mice (*Acomys cahirinus*) and hopping mice (*Notomys alexis*). *Journal of Mammalogy*, 68, 49–57.
- Diaz de Chumaceiro, C. L. (1995). Serendipity or Pseudoserendipity? Unexpected versus desired results. *Journal of Creative Behavior*, 29, 143–147.
- Dickinson, H., Moritz, K., Wintour, E. M., Walker, D. W., & Kett, M. M. (2007). A comparative study of renal function in the desert-adapted spiny mouse and the laboratory-adapted C57BL/6 mouse: Response to dietary salt load. *American Journal of Physiology - Renal Physiology*, 293, F1093–F1098.
- Dickinson, H., & Walker, D. (2007). Managing a colony of spiny mice (*Acomys cahirinus*) for perinatal research. *Australian and New Zealand Council for the Care of Animals in Research and Training (ANZCCART) News*, 20, 4–11.
- Dickinson, H., Walker, D. W., Cullen-McEwen, L., Wintour, E. M., & Moritz, K. (2005). The spiny mouse (*Acomys cahirinus*) completes nephrogenesis before birth. *American Journal of Physiology - Renal Physiology*, 289, F273–F279.
- Driskell, R. R., Lichtenberger, B. M., Hoste, E., Kretzschmar, K., Simons, B. D., Charalambous, M., et al. (2013). Distinct fibroblast lineages determine dermal architecture in skin development and repair. *Nature*, 504, 277–281.
- D'Udine, B., & Alleva, E. (1988). The *Acomys cahirinus* (spiny mouse) as a new model for biological and neurobehavioural studies. *Polish Journal of Pharmacology and Pharmacy*, 40, 525–534.

- Dunoyer, L. A., Seifert, A. W., & Van Cleve, J. (2020). Evolutionary bedfellows: Reconstructing the ancestral state of autotomy and regeneration. *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution*, 336(2), 94–115.
- Ellery, S. J., LaRosa, D. A., Kett, M. M., Della Gatta, P. A., Snow, R. J., Walker, D. W., et al. (2016). Dietary creatine supplementation during pregnancy: A study on the effects of creatine supplementation on creatine homeostasis and renal excretory function in spiny mice. *Amino Acids*, 48, 1819–1830.
- Ellery, S. J., Murthi, P., Davies-Tuck, M. L., Della Gatta, P. A., May, A. K., Kowalski, G. M., et al. (2019). Placental creatine metabolism in cases of placental insufficiency and reduced fetal growth. *Molecular Human Reproduction*, 25, 495–505.
- Ellery, S. J., Murthi, P., Gatta, P. A. D., May, A. K., Davies-Tuck, M. L., Kowalski, G. M., et al. (2020). The effects of early-onset pre-eclampsia on placental creatine metabolism in the third trimester. *International Journal of Molecular Sciences*, 21, 806.
- Emera, D., Romero, R., & Wagner, G. (2012). The evolution of menstruation: A new model for genetic assimilation: Explaining molecular origins of maternal responses to fetal invasiveness. *BioEssays*, 34, 26–35.
- Eming, S. A., Hammerschmidt, M., Krieg, T., & Roers, A. (2009). Interrelation of immunity and tissue repair or regeneration. *Seminars in Cell & Developmental Biology*, 20, 517–527.
- Esposito, M. (2013). Weismann versus morgan revisited: Clashing interpretations on animal regeneration. *Journal of the History of Biology*, 46, 511–541.
- Fabre, P.-H., Hautier, L., Dimitrov, D., & Douzery, E. J. (2012). A glimpse on the pattern of rodent diversification: A phylogenetic approach. *BMC Evolutionary Biology*, 12, 1–19.
- Fernando, W. A., Leininger, E., Simkin, J., Li, N., Malcom, C. A., Sathymoorthi, S., et al. (2011). Wound healing and blastema formation in regenerating digit tips of adult mice. *Developmental Biology*, 350, 301–310.
- Gaire, J., Varholick, J. A., Rana, S., Sunshine, M. D., Doré, S., Barbazuk, W. B., et al. (2021). Spiny mouse (*Acomys*): An emerging research organism for regenerative medicine with applications beyond the skin. *Npj. Regenerative Medicine*, 6, 1–6.
- Gawriluk, T. R., Simkin, J., Hacker, C. K., Kimani, J. M., Kiama, S. G., Ezenwa, V. O., et al. (2020). Complex tissue regeneration in mammals is associated with reduced inflammatory cytokines and an influx of T cells. *Frontiers in Immunology*, 11, 1695.
- Gawriluk, T. R., Simkin, J., Thompson, K. L., Biswas, S. K., Clare-Salzler, Z., Kimani, J. M., et al. (2016). Comparative analysis of ear-hole closure identifies epimorphic regeneration as a discrete trait in mammals. *Nature Communications*, 7, 11164.
- Godwin, J. W., & Brookes, J. P. (2006). Regeneration, tissue injury and the immune response. *Journal of Anatomy*, 209, 423–432.
- Godwin, J. W., Pinto, A. R., & Rosenthal, N. A. (2013). Macrophages are required for adult salamander limb regeneration. *Proceedings of the National Academy of Sciences of the United States of America*, 110, 9415–9420.
- Gould, K. G., Flint, M., & Graham, C. E. (1981). Chimpanzee reproductive senescence: A possible model for evolution of the menopause. *Maturitas*, 3, 157–166.
- Hans, I., Harn, C., Wang, S.-P., Lai, Y.-C., Van Handel, B., Liang, Y.-C., et al. (2021). Symmetry breaking of tissue mechanics in wound induced hair follicle regeneration of laboratory and spiny mice. *Nature Communications*, 12, 1–16.
- Houghton, C. L., Gawriluk, T. R., & Seifert, A. W. (2015). The biology and husbandry of the African spiny mouse (*Acomys cahirinus*) and the research uses of a laboratory Colony. *Journal of the American Association for Laboratory Animal Science*, 55(1), 9–17.
- Hay, E. D., & Fischman, D. A. (1961). Origin of the blastema in regenerating limbs of the newt *Triturus viridescens*. An autoradiographic study using tritiated thymidine to follow cell proliferation and migration. *Developmental Biology*, 3, 26–59.
- Honeycutt, R. L. (2009). Rodents (Rodentia). *The Timetree of Life*, 490–494.

- Hutton, L. C., Ratnayake, U., Shields, A., & Walker, D. W. (2009). Neuropathology and functional deficits in a model of birth asphyxia in the precocial spiny mouse (*Acomys cahirinus*). *Developmental Neuroscience*, 31, 523–535.
- Illingworth, C. M. (1974). Trapped fingers and amputated finger tips in children. *Journal of Pediatric Surgery*, 9, 853–858.
- Ireland, Z., Castillo-Melendez, M., Dickinson, H., Snow, R., & Walker, D. (2011). A maternal diet supplemented with creatine from mid-pregnancy protects the newborn spiny mouse brain from birth hypoxia. *Neuroscience*, 194, 372–379.
- Ireland, Z., Dickinson, H., Fleiss, B., Hutton, L. C., & Walker, D. W. (2010). Behavioural effects of near-term acute fetal hypoxia in a small precocial animal, the spiny mouse (*Acomys cahirinus*). *Neonatology*, 97, 45–51.
- Ito, M., Yang, Z., Andl, T., Cui, C., Kim, N., Millar, S. E., et al. (2007). Wnt-dependent de novo hair follicle regeneration in adult mouse skin after wounding. *Nature*, 447, 316–320.
- Jiang, D., Correa-Gallegos, D., Christ, S., Stefanska, A., Liu, J., Ramesh, P., et al. (2018). Two succeeding fibroblastic lineages drive dermal development and the transition from regeneration to scarring. *Nature Cell Biology*, 20, 422–431.
- Jiang, T. X., Harn, H. I. C., Ou, K. L., Lei, M., & Chuong, C. M. (2019). Comparative regenerative biology of spiny (*Acomys cahirinus*) and laboratory (*Mus musculus*) mouse skin. *Experimental Dermatology*, 28, 442–449.
- Joseph, J., & Dyson, M. (1966). Tissue replacement in the rabbit's ear. *The British Journal of Surgery*, 53, 372–380.
- Kench, J. A., Russell, D. M., Fadok, V. A., Young, S. K., Worthen, G. S., Jones-Carson, J., et al. (1999). Aberrant wound healing and TGF-beta production in the autoimmune-prone MRL/+ mouse. *Clinical Immunology*, 92, 300–310.
- King, M. W., Neff, A. W., & Mescher, A. L. (2012). The developing *Xenopus* limb as a model for studies on the balance between inflammation and regeneration. *Anatomical Record (Hoboken)*, 295, 1552–1561.
- Kingdon, J. (1984). *East African mammals: An atlas of evolution in Africa, volume 2, part B: Hares and rodents*. Chicago: University Of Chicago Press.
- Koopmans, T., van Beijnum, H., Roovers, E. F., Malhotra, D., Tomasso, A., Boeter, J., et al. (2021). Ischemic tolerance and cardiac repair in the African spiny mouse. *NPJ Regenerative Medicine*, 6(1), 1–16.
- Kragl, M., Knapp, D., Nacu, E., Khattak, S., Maden, M., Epperlein, H. H., et al. (2009). Cells keep a memory of their tissue origin during axolotl limb regeneration. *Nature*, 460, 60–65.
- Krogh, A. (1929). The progress of physiology. *American Journal of Physiology-Legacy Content*, 90, 243–251.
- Levi, B. P., & Morrison, S. J. (2008). Stem cells use distinct self-renewal programs at different ages. *Cold Spring Harbor Symposia on Quantitative Biology*, 73, 539–553.
- Lo, D. C., Allen, F., & Brookes, J. P. (1993). Reversal of muscle differentiation during urodele limb regeneration. *Proceedings of the National Academy of Sciences*, 90, 7230–7234.
- Maden, M. (2018). The evolution of regeneration—where does that leave mammals? *International Journal of Developmental Biology*, 62, 369–372.
- Mamrot, J. P. (2019). *Early developmental biology of the spiny mouse (Acomys cahirinus)*. Monash University.
- Mamrot, J., Legaie, R., Ellery, S. J., Wilson, T., Seemann, T., Powell, D. R., et al. (2017). De novo transcriptome assembly for the spiny mouse (*Acomys cahirinus*). *Scientific Reports*, 7, 1–15.
- Manley, G. A. (2012). Evolutionary paths to mammalian cochleae. *Journal of the Association for Research in Otolaryngology*, 13, 733–743.

- Martin, P., & Leibovich, S. J. (2005). Inflammatory cells during wound repair: The good, the bad and the ugly. *Trends in Cell Biology*, 15, 599–607.
- Mascharak, S., Davitt, M. F., Griffin, M., Borrelli, M. R., Moore, A. L., Chen, K., et al. (2021). Preventing Engrailed-1 activation in fibroblasts yields wound regeneration without scarring. *Science*, 372, eaba2374.
- Matias, D. S., Martins, A. R., Casanellas, I., Ova, A. B., Araújo, I. M., Power, D., et al. (2016). Ear wound regeneration in the African spiny mouse *Acomys cahirinus*. *Regeneration*, 3(1), 52–61.
- McKenna, J., Bellofiore, N., Dimitriadis, E., & Temple-Smith, P. (2021). Postpartum ovulation and early pregnancy in the menstruating spiny mouse, *Acomys cahirinus*. *Scientific Reports*, 11, 1–11.
- McKenna, J., Catt, S., Pangestu, M., & Temple-Smith, P. (2020). Successful sperm cryopreservation in Egyptian spiny mice *Acomys cahirinus*. *Reproduction, Fertility and Development*, 32, 1293–1297.
- Mescher, A. L., & Neff, A. W. (2005). Regenerative capacity and the developing immune system. *Advances in Biochemical Engineering/Biotechnology*, 93, 39–66.
- Monaghan, J. R., Athippozhy, A., Seifert, A. W., Putta, S., Stromberg, A. J., Maden, M., et al. (2012). Gene expression patterns specific to the regenerating limb of the Mexican axolotl. *Biology Open*, 1, 937–948.
- Monaghan, J. R., & Maden, M. (2013). Cellular plasticity during vertebrate appendage regeneration. *Current Topics in Microbiology and Immunology*, 367, 53–74.
- Montandon, S. A., Tzika, A. C., Martins, A. F., Chopard, B., & Milinkovitch, M. C. (2014). Two waves of anisotropic growth generate enlarged follicles in the spiny mouse. *EvoDevo*, 5, 1–13.
- Mori, R., Shaw, T. J., & Martin, P. (2008). Molecular mechanisms linking wound inflammation and fibrosis: Knockdown of osteopontin leads to rapid repair and reduced scarring. *The Journal of Experimental Medicine*, 205, 43–51.
- Muneoka, K., Allan, C. H., Yang, X., Lee, J., & Han, M. (2008). Mammalian regeneration and regenerative medicine. *Birth Defects Research. Part C, Embryo Today*, 84, 265–280.
- Needham, A. E. (1952). *Regeneration and wound-healing*. Methuen & Co.
- Neufeld, D. A. (1980). Partial blastema formation after amputation in adult mice. *Journal of Experimental Zoology*, 212, 31–36.
- Neufeld, D. A., & Zhao, W. (1995). Bone regrowth after digit tip amputation in mice is equivalent in adults and neonates. *Wound Repair and Regeneration*, 3, 461–466.
- Oberpriller, J. O., & Oberpriller, J. C. (1974). Response of the adult newt ventricle to injury. *Journal of Experimental Zoology*, 187, 249–259.
- Okamura, D. M., Brewer, C. M., Wakenight, P., Bahrami, N., Bernardi, K., Tran, A., et al. (2018). Scarless repair of acute and chronic kidney injury in African spiny mice (*Acomys cahirinus*). *bioRxiv*, 315069.
- Oosterhuis, W., Mooren, P., Charles, R., & Lamers, W. (1984). Perinatal development of the lung in rat and spiny mouse: Its relation to altricial and precocial timing of birth. *Neonatology*, 45, 236–243.
- Pajcini, K. V., Corbel, S. Y., Sage, J., Pomerantz, J. H., & Blau, H. M. (2010). Transient inactivation of Rb and ARF yields regenerative cells from postmitotic mammalian muscle. *Cell Stem Cell*, 7, 198–213.
- Pasco, R., Gardner, D. K., & Walker, D. W. (2012). A superovulation protocol for the spiny mouse (*Acomys cahirinus*). *Reproduction, Fertility and Development*, 24(8), 1117–1122.
- Peitz, B. (1981). The oestrous cycle of the spiny mouse (*Acomys cahirinus*). *Reproduction*, 61, 453–459.
- Peitz, B., Foreman, D., & Schmitt, M. (1979). The reproductive tract of the male spiny mouse (*Acomys cahirinus*) and coagulation studies with other species. *Reproduction*, 57, 183–188.

- Peng, H., Shindo, K., Donahue, R. R., Gao, E., Ahern, B. M., Levitan, B. M., et al. (2021). Adult spiny mice (*Acomys*) exhibit endogenous cardiac recovery in response to myocardial infarction. *NPJ Regenerative Medicine*, 6(1), 1–15.
- Petrie, T. A., Strand, N. S., Tsung-Yang, C., Rabinowitz, J. S., & Moon, R. T. (2014). Macrophages modulate adult zebrafish tail fin regeneration. *Development*, 141, 2581–2591.
- Phan, Q. M., Fine, G. M., Salz, L., Herrera, G. G., Wildman, B., Driskell, I. M., et al. (2020). Lefl expression in fibroblasts maintains developmental potential in adult skin to regenerate wounds. *eLife*, 9, e60066.
- Poss, K. D., Wilson, L. G., & Keating, M. T. (2002). Heart regeneration in zebrafish. *Science*, 298, 2188–2190.
- Qi, Y., Dasa, O., Maden, M., Vohra, R., Batra, A., Walter, G., et al. (2021). Functional heart recovery in an adult mammal, the spiny mouse. *International Journal of Cardiology*.
- Rasweiler, J. J. (1991). Spontaneous decidual reactions and menstruation in the black mastiff bat, *Molossus ater*. *American Journal of Anatomy*, 191, 1–22.
- Rasweiler, J. J., & De Bonilla, H. (1992). Menstruation in short-tailed fruit bats (*Carollia* spp.). *Reproduction*, 95, 231–248.
- Ratnayake, U., Quinn, T., Daruwalla, K., Dickinson, H., & Walker, D. W. (2014). Understanding the behavioural phenotype of the precocial spiny mouse. *Behavioural Brain Research*, 275, 62–71.
- Reines, B., Cheng, L. I., & Matzinger, P. (2009). Unexpected regeneration in middle-aged mice. *Rejuvenation Research*, 12, 45–52.
- Robinson, S. R., & Smotherman, W. P. (1992). Behavioral response of altricial and precocial rodent fetuses to acute umbilical cord compression. *Behavioral and Neural Biology*, 57, 93–102.
- Rognoni, E., Gomez, C., Pisco, A. O., Rawlins, E. L., Simons, B. D., Watt, F. M., et al. (2016). Inhibition of β -catenin signalling in dermal fibroblasts enhances hair follicle regeneration during wound healing. *Development*, 143, 2522–2535.
- Sandoval-Guzman, T., Wang, H., Khattak, S., Schuez, M., Roensch, K., Nacu, E., et al. (2014). Fundamental differences in dedifferentiation and stem cell recruitment during skeletal muscle regeneration in two salamander species. *Cell Stem Cell*, 14, 174–187.
- Satoh, A., Hirata, A., & Makanae, A. (2012). Collagen reconstitution is inversely correlated with induction of limb regeneration in *Ambystoma mexicanum*. *Zoological Science*, 29, 191–197.
- Satoh, A., Hirata, A., & Satou, Y. (2011). Blastema induction in aneurogenic state and *Prrx-1* regulation by MMPs and FGFs in *Ambystoma mexicanum* limb regeneration. *Developmental Biology*, 355, 263–274.
- Saxena, S., Vekaria, H., Sullivan, P. G., & Seifert, A. W. (2019). Connective tissue fibroblasts from highly regenerative mammals are refractory to ROS-induced cellular senescence. *Nature Communications*, 10, 1–16.
- Seifert, A. W., Kiama, S. G., Seifert, M. G., Goheen, J. R., Palmer, T. M., & Maden, M. (2012). Skin shedding and tissue regeneration in African spiny mice (*Acomys*). *Nature*, 489, 561–565.
- Seifert, A. W., Monaghan, J. R., Voss, S. R., & Maden, M. (2012). Skin regeneration in adult axolotls: A blueprint for scar-free healing in vertebrates. *PLoS One*, 7, e32875.
- Sekulić, S., Božić, K., Božić, A., Borota, J., & Čulić, M. (2006). Precocial rodents as new experimental model to study the effects of altered gravitational conditions on fetal development. *Microgravity Science and Technology*, 18, 223–225.
- Shafir, E., Ziv, E., & Kalman, R. (2006). Nutritionally induced diabetes in desert rodents as models of type 2 diabetes: *Acomys cahirinus* (spiny mice) and *Psammomys obesus* (desert gerbil). *ILAR Journal*, 47, 212–224.
- Shargal, E., Rath-Wolfson, L., Kronfeld, N., & Dayan, T. (1999). Ecological and histological aspects of tail loss in spiny mice (Rodentia: Muridae, *Acomys*) with a review of its occurrence in rodents. *Journal of Zoology*, 249, 187–193.
- Simkin, J., Gawriluk, T. R., Gensel, J. C., & Seifert, A. W. (2017). Macrophages are necessary for epimorphic regeneration in African spiny mice. *eLife*, 6.

- Simkin, J., & Seifert, A. W. (2018). Concise review: Translating regenerative biology into clinically relevant therapies: Are we on the right path? *Stem Cells Translational Medicine*, 7, 220–231.
- Singer, M., Weckesser, E. C., Géraudie, J., Maier, C. E., & Singer, J. (1987). Open finger tip healing and replacement after distal amputation in rhesus monkey with comparison to limb regeneration in lower vertebrates. *Anatomy and Embryology*, 177, 29–36.
- Sinigaglia, C., & Averof, M. (2019). The multifaceted role of nerves in animal regeneration. *Current Opinion in Genetics & Development*, 57, 98–105.
- Steppan, S. J., & Schenk, J. J. (2017). Muroid rodent phylogenetics: 900-species tree reveals increasing diversification rates. *PLoS One*, 12, e0183070.
- Stocum, D. L. (2011). The role of peripheral nerves in urodele limb regeneration. *European Journal of Neuroscience*, 34, 908–916.
- Streeter, K. A., Sunshine, M. D., Brant, J. O., Sandoval, A. G., Maden, M., & Fuller, D. D. (2020). Molecular and histologic outcomes following spinal cord injury in spiny mice, *Acomys cahirinus*. *Journal of Comparative Neurology*, 528, 1535–1547.
- Tanaka, E. M., Gann, A. A., Gates, P. B., & Brookes, J. P. (1997). Newt myotubes reenter the cell cycle by phosphorylation of the retinoblastoma protein. *The Journal of Cell Biology*, 136, 155–165.
- Temple-Smith, P. (1973). *The seasonal breeding biology of the platypus, Ornithorhynchus anatinus, with special reference to the male*. Department of Zoology: Australian National University.
- Temple-Smith, P. D. (2003). 14 integrating reproductive sciences into recovery programmes for declining and extinct marsupial populations. In *Reproductive science and integrated conservation* (p. 211). Cambridge University Press.
- Temple-Smith, P., & Grant, T. (2001). Uncertain breeding: A short history of reproduction in monotremes. *Reproduction, Fertility and Development*, 13, 487–497.
- Temple-Smith, P., Southwick, G., Yates, C., Trounson, A., & De Kretser, D. (1985). Human pregnancy by in vitro fertilization (IVF) using sperm aspirated from the epididymis. *Journal of In Vitro Fertilization and Embryo Transfer*, 2, 119–122.
- Van der Horst, C. (1946). Some remarks on the biology of reproduction in the female of *Elephantulus*, the holy animal of set. *Transactions of the Royal Society of South Africa*, 31, 181–199.
- Verma, R., Holland, M., Temple-Smith, P., & Verma, P. (2012). Inducing pluripotency in somatic cells from the snow leopard (*Panthera uncia*), an endangered felid. *Theriogenology*, 77, 220–228, (e222).
- Vorontsova, M. a. L., LD (1960). *Asexual propagation and regeneration*. New York: Pergamon Press.
- Wang, X., Catt, S., Pangestu, M., & Temple-Smith, P. (2009). Live offspring from vitrified blastocysts derived from fresh and cryopreserved ovarian tissue grafts of adult mice. *Reproduction*, 138, 527.
- Wang, X., Hsi, T. C., Guerrero-Juarez, C. F., Pham, K., Cho, K., McCusker, C. D., et al. (2015). Principles and mechanisms of regeneration in the mouse model for wound-induced hair follicle neogenesis. *Regeneration (Oxf)*, 2, 169–181.
- Wang, W., Hu, C.-K., Zeng, A., Alegre, D., Hu, D., Gotting, K., et al. (2020). Changes in regeneration-responsive enhancers shape regenerative capacities in vertebrates. *Science*, 369.
- Williams-Boyce, P. K., & Daniel, J. C., Jr. (1980). Regeneration of rabbit ear tissue. *The Journal of Experimental Zoology*, 212, 243–253.
- Williams-Boyce, P. K., & Daniel, J. C., Jr. (1986). Comparison of ear tissue regeneration in mammals. *Journal of Anatomy*, 149, 55–63.
- Wilson, E. O. (1989). The coming pluralization of biology and the stewardship of systematics. *Bioscience*, 39, 242–245.
- Zakaria, R., Al Rahbi, B., Ahmad, A. H., Said, R. M., Othman, Z., Azman, K. F., et al. (2019). Menopause rodent models: Suitability for cognitive aging research. *International Medical Journal*, 26, 450–452.